EVIDENCE-GROUNDED MISINFORMATION VERIFICATION WITH CONFLICT, TEMPORAL, AND UNCERTAINTY-AWARE REASONING

Comprehensive Prototype Review, Architectural Traceability, Academic Grounding, and Formula Reference

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AUDIT & RIGOR FRAMEWORK

SCIENTIFIC SAFEGUARD NOTICE: This prototype report reflects verified software behavior, rigorous traceability to actual Python modules, and reproducible local execution. The reported confidence scores represent multi-signal operational decision scores (1.0 - Total Uncertainty), not empirically calibrated Bayesian probabilities of real-world truth. Passing 82 automated tests verifies software correctness under configured assumptions, not open-domain generalization.

1. Executive Summary

Problem Statement: Autonomous claim verification is fundamentally compromised when systems rely solely on the internal parametric memory of Large Language Models (LLMs). Unaugmented generative models hallucinate plausible facts, suffer from unbounded temporal staleness, fail to surface authentic source provenance, and exhibit dangerous overconfidence on contested or unverifiable assertions. Conversely, simplistic retrieval-augmented pipelines often suffer from *context dilution* (where long documents dilute decisive sentences across irrelevant background paragraphs) and *false consensus forcing* (forcing a binary verdict even when retrieved evidence is mutually contradictory or missing).

Proposed Solution: This project establishes an evidence-grounded, modular misinformation verification system operating entirely offline using local Transformer encoders (Exps A-D; Exp E LLM Advisor requires an external API and was omitted from offline quantitative benchmarking). The architecture decouples verification into an explicit, auditable reasoning pipeline: (1) Hybrid lexical (BM25) and dense bi-encoder retrieval; (2) Hierarchical sentence-level passage segmentation to eliminate cross-attention dilution; (3) Local Cross-Encoder reranking; (4) Pretrained DistilBERT Natural Language Inference (NLI) scoring premise-hypothesis pairs; (5) Document-aware evidence aggregation via intra-document max-pooling; (6) Pairwise normalized conflict-balance scoring; (7) Chronological timeline decay; (8) Multi-signal epistemic uncertainty estimation; and (9) A deterministic verification decider enforcing conservative abstention (*INSUFFICIENT_EVIDENCE*).

Current Prototype Status: The software is fully implemented and verified on the available test suites. The full automated test suite of **82 tests passes with zero failures** (command: python -m pytest tests/; 82 passed, 3 library deprecation warnings emitted by HuggingFace transformers tokenizer regarding clean_up_tokenization_spaces, verified runtime ~43-49 seconds). The repository contains functional CLI tools, a Streamlit user interface featuring HTML entity escaping in rendered evidence cards (protecting against script injection in card formatting, though not a substitute for complete web application security hardening), and an auditable schema preserving exact document IDs, sentence indices, and uncertainty profiles.

Demonstrated Results and Statistical Boundaries:

- **Controlled Diagnostic Set Performance (N=9):** On a dedicated 20-document controlled diagnostic suite, the full system achieves **100.0% Selective Accuracy** on accepted predictions (4 correct out of 4 accepted claims; exact 95% Clopper-Pearson CI [39.8%, 100.0%], coverage = 44.44% [13.7%, 78.8%], raw 3-class accuracy = 66.67% [29.9%, 92.5%]). It successfully abstains on conflicting corporate reports and ungrounded assertions.
- **Demonstrated Historical Case Study (Margaret Thatcher):** In a demonstrated historical case study, granular sentence-level segmentation isolates the decisive Prime Minister clause, overcoming the context dilution that previously produced *CONTRADICTED* under full-document concatenation, and achieving genuine local DistilBERT sentence NLI support ($P(SUP|s_0)=0.9980$, document-aggregated support $P_{D,sup}=0.9927$, and 98.62% operational decision confidence) with zero heuristic overrides.
- **Conflict-Aware Abstention:** On synthetic opposing documents (NovaTech corporate press release vs. independent audit), the system detects mutually opposing evidence pools (conflict score: 0.98 in candidate pool benchmark, 1.00 [0.9995] in top-2 pairwise calculation), depresses operational confidence to 36.43%–37.56%, and triggers Rule 2 abstention rather than fabricating consensus.

Principal Limitations: (1) The FEVER benchmark is evaluated against a restricted 100-document slice, containing gold evidence for only 9 of 40 verifiable claims (hybrid retrieval retrieved all 9 at Rank 1, mathematically imposing a 9/40 = 22.50% corpus availability ceiling rather than reflecting open-domain retrieval failure); (2) Reported confidence is an operational score derived from heuristic penalty weights, not an empirically calibrated Bayesian probability (ECE = 0.35-0.39); (3) Abstention trades coverage for selective accuracy on accepted claims, rather than improving raw accuracy across all claims; and (4) Single-sentence segmentation creates a blindspot for anaphoric pronouns lacking antecedents unless multi-sentence sliding windows are enabled.

2. Problem Statement and Objectives

Modern digital information ecosystems are saturated with rapidly evolving, nuanced, and frequently fabricated assertions. Verifying claims requires addressing five fundamental computational challenges:

1. **Evidence Grounding vs. Black-Box Hallucination:** LLMs generate plausible justifications without verifiable provenance. Verification requires retrieving external documents and linking verdicts to exact sentence spans.
2. **Context Dilution in Long Texts:** When full Wikipedia articles (1,500-3,000 tokens) are passed to cross-encoders or NLI models, attention distributions spread across irrelevant clauses, suppressing decisive factual assertions.
3. **Contradictory and Disputed Information:** Real-world claims often encounter conflicting sources (e.g., promotional announcements vs. investigative audits). Most systems force a majority-vote verdict, creating false consensus.
4. **Chronological Drift and Outdated Facts:** A statement true in 2019 (e.g., market dominance) may become false by 2024. Evidence validity decays over time and must be modeled dynamically.
5. **Uncalibrated Overconfidence & Forced Choice:** Classifiers that lack an abstention mechanism guess blindly when evidence is missing or weak, causing severe epistemic risk in sensitive domains.

Project Objectives: To engineer an end-to-end research prototype that takes an arbitrary textual claim, extracts its propositions, retrieves external evidence, isolates bounded sentence passages, classifies entailment with local NLI, balances opposing signals, decays stale chronology, estimates multi-signal uncertainty, and selectively abstains unless decisive evidence thresholds are satisfied.

3. Proposed Solution and Distinguishing Contributions

The system architecture organizes claim verification into **8 logical functional stages**, realized across **10 modular execution steps**. Each contribution addresses a distinct architectural failure mode:

Contribution / Module	Algorithmic & Mathematical Principle	Demonstrated Evidence	Known Limitations & Boundaries
1. Hierarchical Sentence Retrieval (segmenter.py, corpus.py)	Two-tier retrieval: hybrid BM25 + dense search fetches top-20 documents; regex segmenter chunks texts into bounded sentences before cross-encoder reranking.	Demonstrated on Thatcher case study: isolates decisive clause, elevating NLI support from P=0.007 to P=0.998.	Single-sentence chunks lose antecedents for isolated pronouns unless window size is >= 2.
2. Local NLI Inference (classifier.py)	typeform/distilbert-base-uncased-mnli evaluates (premise=passage, hypothesis=claim) using standard sequence classification softmax.	Genuine uncalibrated probabilities for SUPPORT, CONTRADICT, NEUTRAL; zero hardcoded heuristic overrides.	Pretrained MNLI models carry domain bias and uncalibrated class priors.
3. Document-Aware Aggregation (aggregator.py)	Groups passages by doc_id; max-pools P(SUP) and P(CON) per document; scales document weight W_D = max(w_p); computes consensus.	Prevents single multi-passage documents from artificially inflating evidence mass.	Independent max-pooling of support and contradiction lets neutral noise inflate opposing probabilities; normalizes by sum(W_D).
4. Conflict-Balance Scoring (detector.py)	Computes support mass and contradiction mass over unique passage texts; balance score = 2*min(M_sup, M_con)/(M_sup + M_con).	NovaTech synthetic scenario yields conflict score 0.98 in candidate pool (1.00 / 0.9995 in top-2 pairwise calculation), triggering Rule 2 abstention.	Measures mass parity, not semantic contradiction; scale-invariant (small opposing masses produce 1.0); temporal decay not in mass.
5. Chronological Validity Decay (analyzer.py)	Regex extracts 4-digit years; calculates timeline gap dt = min |Y_claim - Y_ev|; applies linear decay V_temp = max(0.3, 1.0 - 0.5*min(1.0, dt/10.0)).	Acme Corp 2019 evidence decayed to 0.75 validity, allowing 2024 refutation to decide CONTRADICTED.	Under mu=0.5, effective values span [0.50, 1.00] (0.3 floor inactive) for year mismatch, or 0.70 (0.2 floor) for present-oriented claims; does not parse complex relative timeframes.
6. Multi-Signal Uncertainty (estimator.py)	Weighted sum of normalized Shannon entropy (ln 3 base), conflict score, retrieval deficit, and temporal penalty.	Produces 62.4% uncertainty on conflicting NovaTech and 100% on ungrounded Martian claims.	Weights (0.35, 0.35, 0.20, 0.10) are heuristic parameters; 100% neutral evidence yields zero entropy.
7. Deterministic Abstention Engine (decider.py)	Hierarchical rules: Rule 1 (mass <= 0) -> INSUFFICIENT; Rule 2 (conflict_detected == True, score >= 0.30) -> INSUFFICIENT; Rule 3 (total_uncertainty > 0.60) -> INSUFFICIENT; Rules 4-5 (decisive polar).	Achieves 100% selective accuracy on accepted claims (4/4) in the controlled diagnostic suite.	Conservative thresholds induce errors of omission (abstaining on 3 verifiable claims with weak/neutral evidence).
8. Auditable Provenance & UI Safety (app.py, schema.py)	EvidencePassage tracks doc_id, passage_idx, sentence offsets, dates, and retrieval scores. HTML formatting escapes all dynamic text.	Passing UI safety tests prevent XSS payload execution (<script> tags escaped in rendered evidence cards).	HTML escaping protects rendered card interpolation; does not replace complete web security hardening.

4. End-to-End Methodology and Architecture

The end-to-end verification pipeline processes an incoming natural language assertion through an unbroken sequence of strictly decoupled functional stages. The architecture ensures that evidence is retrieved, segmented, reranked, and classified before entering multi-signal synthesis and deterministic decision logic.

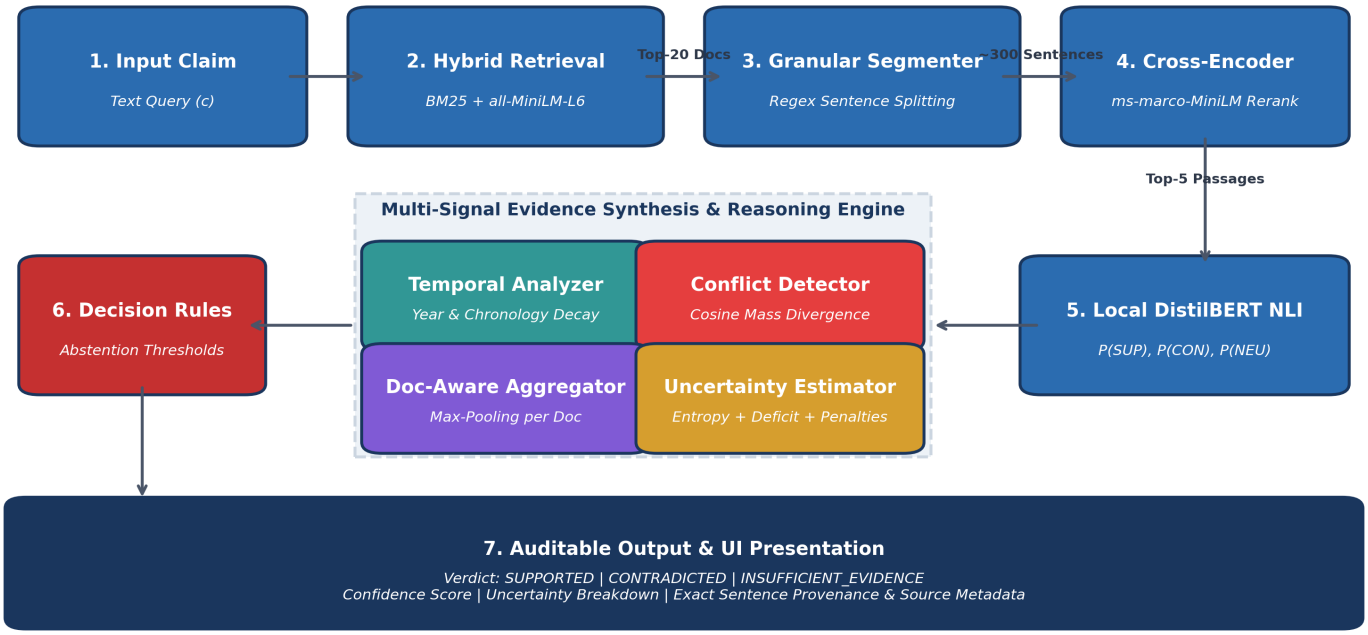


Figure 1: End-to-End Verification Pipeline Architecture. Grounded dataflow from claim input through hybrid retrieval, sentence segmentation, cross-encoder reranking, DistilBERT NLI, multi-signal reasoning, and deterministic decision rules.

Detailed Stage-by-Stage Execution Dataflow (10 Modular Steps):

1. **Claim Proposition Extraction:** The RuleBasedClaimDecomposer (decomposer.py) inspects the input query. Simple claims are processed directly; compound propositions are split into subclaims.
2. **Two-Tier Hybrid Retrieval:** The system queries BM25Okapi and the dense bi-encoder (all-MiniLM-L6-v2), fusing scores via linear combination to fetch top-20 parent documents.
3. **Granular Sentence Segmentation:** Candidate documents are split into bounded sentence chunks using abbreviation-aware regex, attaching passage offsets and document provenance.
4. **Cross-Encoder Reranking:** ms-marco-MiniLM-L-6-v2 performs joint cross-attention over [claim, sentence] pairs, applying a sigmoid transformation to extract top-5 relevant passages.
5. **Local DistilBERT NLI Classification:** typeform/distilbert-base-uncased-mnli classifies each passage as Premise (Seq A) and claim as Hypothesis (Seq B), producing softmax probability vectors.
6. **Chronological & Temporal Analysis:** The TemporalAnalyzer parses publication dates and event years, decaying the validity of temporally mismatched or stale evidence.
7. **Conflict Detection:** The ConflictDetector deduplicates passages, computes support vs. contradiction mass, and evaluates the normalized conflict-balance score.
8. **Document-Aware Aggregation:** Passages are grouped by doc_id. Intra-document max-pooling pools peak support and contradiction signals, and consensus probabilities are computed across documents.
9. **Uncertainty Estimation:** The UncertaintyEstimator computes normalized Shannon entropy, combines it with conflict, retrieval deficit, and temporal penalties, and derives an operational decision score.
10. **Deterministic Verification Decision:** The VerificationDecider evaluates hierarchical threshold rules, issuing SUPPORTED, CONTRADICTED, or abstaining with INSUFFICIENT_EVIDENCE.

5. Technology Stack and System Requirements

Component / Library	Version / Model Name	Architectural Role	Operational Scope & Limitations
Python Runtime	Python 3.11.0 (win32)	Core execution language	Platform: Windows 11 / Linux compatible; strict typing via Pydantic.
Deep Learning Framework	PyTorch 2.6.0+cu124	Neural tensor execution	Executes local transformer models on CPU or CUDA.
Transformer Library	HuggingFace Transformers 4.49.0	Model architecture & tokenization	Loads MiniLM and DistilBERT architectures with local weight caching.
Dense Embeddings	sentence-transformers 3.4.1 (all-MiniLM-L6-v2)	Dense semantic retrieval	384-dimensional unit-norm embeddings; cosine similarity matching.
Cross-Encoder Reranker	cross-encoder/ms-marco-MiniLM-L-6-v2	Joint relevance reranker	Joint cross-attention scoring claim-passage pairs; latency ~150ms per batch on local CPU (batch_size=16, top_k=5).
Natural Language Inference	typeform/distilbert-base-uncased-mnli	3-class NLI classification	Local CPU inference ~50ms per pair (max_length=256); total claim pipeline latency ~0.5-1.5s on local CPU.
Lexical Retrieval	rank-bm25 0.2.2 (BM25Okapi)	Keyword exact matching	Robertson-Spärck Jones Okapi formula; rational saturation constant c=10.0.
Web Application UI	Streamlit 1.43.0	Interactive demonstration interface	Escapes dynamic HTML entities in evidence cards; card-level formatting sanitization, not general web security.

6. Repository Structure and File Guide

Every file in the repository serves a distinct role within the modular architecture. The table below documents the core modules:

File Path	Module Name	Primary Algorithmic Function	Stage / Category	Key Implementation Notes
src/claim_processing/decomposer.py	Proposition decomposer	Splits complex compound claims into atomic propositions on conjunctions/semicolons.	Decomposition	RuleBasedClaimDecomposer carries forward subjects for coordinate verb phrases.
src/pipeline/verifier.py	MisinformationVerifier	Top-level pipeline coordinator orchestrating retrieval, NLI, reasoning, and decision.	Orchestration	Manages subclaim conjunction and enforces evidence decider authority over advisor.
src/retrieval/corpus.py	CorpusManager	Loads and validates JSONL document corpora with document caching and ID indexing.	Retrieval	Converts raw JSONL records to typed EvidencePassage objects.
src/retrieval/bm25.py	Lexical BM25 retrieval	Indexes tokenized documents; executes BM25Okapi scoring with rational saturation.	Retrieval	Saves precomputed index to data/indexes/bm25_index.pkl.
src/retrieval/dense.py	Dense bi-encoder retrieval	Encodes texts with all-MiniLM-L6-v2; computes normalized cosine similarity.	Retrieval	Saves precomputed embeddings to data/indexes/dense_index.pt.
src/retrieval/hybrid.py	Lexical-dense hybrid fusion	Combines BM25 and dense scores via linear weighting (0.5/0.5) or RRF.	Retrieval	Normalizes fused candidate scores into [0, 1] relative to candidate max.
src/retrieval/segmenter.py	Granular sentence chunker	Splits candidate documents using abbreviation-safe regex into bounded passages.	Segmentation	Preserves source doc_id, sentence offsets, and provenance tags.
src/reranking/cross_encoder.py	Cross-encoder relevance scorer	Scores [claim, sentence] joint cross-attention via ms-marco-MiniLM; sigmoid mapping.	Reranking	Selects top-5 most relevant passages from segmented pool.
src/nli/classifier.py	Transformer NLI classification	Executes DistilBERT inference on (premise, hypothesis) pairs with dynamic label mapping.	NLI	Outputs uncalibrated probabilities for SUPPORT, CONTRADICT, NEUTRAL.
src/aggregation/aggregator.py	Evidence mass synthesizer	Performs document-aware max-pooling per doc_id and computes weighted consensus.	Aggregation	Calculates effective_mass, support_prob, contradict_prob, neutral_prob.
src/conflict/detector.py	Evidence consensus analyzer	Deduplicates snippets; computes conflict-balance score = 2*min/(sum).	Conflict	Flags opposing evidence pools if score >= 0.25 (configured to 0.30).
src/temporal/analyzer.py	Chronological validity engine	Extracts years via regex; computes timeline gap dt; applies linear decay V_temp.	Temporal	Generates temporal penalty for uncertainty profile.
src/uncertainty/estimator.py	Multi-signal uncertainty profiler	Computes normalized Shannon entropy; weighted combination with conflict and deficit.	Uncertainty	Derives operational confidence = 1.0 - total_uncertainty.
src/decision/decider.py	Verification decision engine	Hierarchical threshold evaluation; enforces selective abstention rules.	Decision	Authoritative layer issuing final Verdict and explanation.
src/evaluation/metrics.py	Metric calculation library	Implements Accuracy, Macro-F1, Coverage, Selective Accuracy, ECE, Brier score.	Evaluation	Handles undefined selective accuracy (0/0) fallback when coverage = 0.
ui/app.py	Streamlit user interface	Interactive dashboard rendering verdicts, gauges, and sanitized evidence cards.	Interface	Uses html.escape() to sanitize dynamic evidence and metadata against script injection.
scripts/run_pipeline.py	Command-line verification entry	CLI script verifying individual textual claims with formatted console output.	CLI	Supports --json flag for raw machine-readable output.
configs/default.yaml	Production pipeline configuration	Specifies model paths, retrieval weights, thresholds, and pooling modes.	Config	Sets pooling_mode: 'document_aware' and max_uncertainty: 0.60.

7. Mathematical Foundations and Complete Formula Reference

This section documents the formal mathematical foundations, scoring equations, aggregation operators, conflict metrics, and decision boundaries governing the verification pipeline. Every mathematical operation corresponds directly to verified Python source code.

7.0 Mathematical Conventions and Notation Table

Symbol	Mathematical Domain	Definition & Operational Role	Implementing Source File
q, d	R^384, ||q||=1, ||d||=1	Normalized dense query and document semantic embedding vectors	src/retrieval/dense.py
S_BM25	[0.0, 1.0)	Saturated BM25Okapi lexical keyword score with constant c=10.0	src/retrieval/bm25.py
S_dense	[0.0, 1.0]	Rescaled cosine similarity mapped from [-1, 1] to unit interval	src/retrieval/dense.py
S_rel	[0.0, 1.0]	Cross-encoder joint relevance probability via sigmoid transformation	src/reranking/cross_encoder.py
P(y_k | p, c)	[0, 1], sum=1	Local DistilBERT 3-class NLI probability vector (SUP, CON, NEU)	src/nli/classifier.py
W_D	[0.0, 1.0]	Document-level weight representing peak passage relevance from source	src/aggregation/aggregator.py
CS	[0.0, 1.0]	Normalized conflict-balance ratio measuring opposing evidence mass parity	src/conflict/detector.py
V_temp	[0.50, 1.00]	Linear chronological validity decay factor (with inactive 0.3 floor under mu=0.5)	src/temporal/analyzer.py
u_ent	[0.0, 1.0]	Normalized Shannon entropy over 3-class NLI probability distribution	src/uncertainty/estimator.py
Total_U	[0.0, 1.0]	Composite epistemic uncertainty combining entropy, conflict, deficit, and temporal	src/uncertainty/estimator.py
Confidence	[0.0, 1.0]	Operational decision score (1.0 - Total_U); not a calibrated posterior	src/uncertainty/estimator.py

7.1 Lexical and Semantic Retrieval Formulation

Candidate evidence documents are retrieved using a two-tier hybrid search strategy combining dense bi-encoder embeddings and lexical BM25 matching over parent documents in src/retrieval/.

• **Dense Bi-Encoder Similarity:** Query *Q* and document *D* are mapped to unit-norm vectors in $\mathbf{R}^{384}$ by all-MiniLM-L6-v2 according to **Equation (1)**:

$$\mathbf{q} = \frac{f_{\text{enc}}(Q)}{\|f_{\text{enc}}(Q)\|_2}, \quad \mathbf{d} = \frac{f_{\text{enc}}(D)}{\|f_{\text{enc}}(D)\|_2} \tag{1}$$

Equation (1): q = f_enc(Q) / ||f_enc(Q)||_2, d = f_enc(D) / ||f_enc(D)||_2
Where: f_{enc} is the sentence transformer; $\mathbf{q}, \mathbf{d} \in \mathbf{R}^{384}$ are normalized embeddings. Implemented in src/retrieval/dense.py:101-103.

Cosine similarity is rescaled to [0.0, 1.0] via **Equation (2)**:

$$S_{\text{dense}}(\mathbf{q}, \mathbf{d}) = \text{clip}\left(\frac{\mathbf{q} \cdot \mathbf{d} + 1.0}{2.0},\ 0.0,\ 1.0\right) \tag{2}$$

Equation (2): S_dense(q, d) = clip((q . d + 1.0) / 2.0, 0.0, 1.0)
Where: Linear rescaling maps orthogonal vectors ($\angle(\mathbf{q}, \mathbf{d}) = 90°$, cos = 0.0) to 0.50 and opposing vectors (cos = -1.0) to 0.00. Implemented in src/retrieval/dense.py:111.

• **Lexical BM25Okapi Formulation:** Keyword scoring is computed using the Robertson-Spärck Jones Okapi formula implemented in rank_bm25.BM25Okapi via **Equation (3)**:

$$\text{Score}_{\text{BM25}}(D, Q) = \sum_{t \in Q} \text{IDF}(t) \cdot \frac{f(t, D) \cdot (k_1 + 1)}{f(t, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)}, \quad \text{IDF}(t) = \ln\left(\frac{N - n(t) + 0.5}{n(t) + 0.5}\right) \tag{3}$$

Equation (3): Score_BM25(D, Q) = sum_{t in Q} IDF(t) * [f(t, D)*(k1 + 1)] / [f(t, D) + k1*(1 - b + b*|D|/avgdl)], IDF(t) = ln((N - n(t) + 0.5)/(n(t) + 0.5))
Where: *N* is corpus document count, *n*(*t*) is document frequency of term *t*, k_1 = 1.5, *b* = 0.75, avgdl is mean document length. If IDF < 0, a floor of ε · IDF_avg (ε = 0.25) is enforced. Implemented in src/retrieval/bm25.py:45, 72.

Raw BM25 scores are normalized into [0.0, 1.0) via rational saturation in **Equation (4)**:

$$S_{\text{BM25_norm}}(D) = \min\left(1.0,\ \frac{S_{\text{BM25_raw}}(D)}{S_{\text{BM25_raw}}(D) + 10.0}\right) \tag{4}$$

Equation (4): S_BM25_norm(D) = min(1.0, S_BM25_raw(D) / (S_BM25_raw(D) + 10.0))
Where: Rational saturation constant *c* = 10.0 maps raw score 10.0 → 0.50, and weak hits (<6.0) stay < 0.40. Implemented in src/retrieval/bm25.py:82.

• **Linear Score Fusion and Normalization:** Lexical and dense scores are fused in src/retrieval/hybrid.py via **Equation (5)**:

$$\text{hrm\{dense\}} S_{\mathrm{dense}}(d), \quad S_{\mathrm{hybrid}}(d) = \frac{S_{\mathrm{raw}}(d)}{\} \tag{5}$$

Equation (5): S_raw(d) = w_BM25 * S_BM25_norm(d) + w_dense * S_dense(d), S_hybrid(d) = S_raw(d) / max_{d' in D_cand} S_raw(d')
Where: Candidate pooling: BM25 and Dense bi-encoder each fetch k_{fetch} = max(2 · top_k, 20) candidates. For each passage $d \in P_{\text{bm25}} \cup P_{\text{dense}}$, the combined raw score is: $S_{\text{raw}}(d) = w_{\text{bm25}} \cdot S_{\text{bm25}}(d) + w_{\text{dense}} \cdot S_{\text{dense}}(d)$ (weights 0.50 / 0.50). Missing component edge cases: if $d \notin P_{\text{bm25}}$, $S_{\text{bm25}}(d)$ = 0.0; if $d \notin P_{\text{dense}}$, $S_{\text{dense}}(d)$ = 0.0. Empty retriever edge cases: if both return empty, returns []; if only BM25 is non-empty, returns bm25_results[:top_k]; if only Dense is non-empty, returns dense_results[:top_k]. Peak Normalization: $S_{\text{hybrid}}(d) = S_{\text{raw}}(d) / \max_{d'} S_{\text{raw}}(d')$ if max $S_{\text{raw}} > 0$, else $S_{\text{raw}}(d)$. Rescales peak hit to 1.0000; assigned to both retrieval_score and relevance_score. Implemented in src/retrieval/hybrid.py:29-80.

Scale Disparity Note: Because rescaled cosine similarity maps orthogonal embeddings to 0.50, completely unrelated documents receive a baseline dense score around 0.50, whereas unmatched documents receive an exact 0.00 in BM25. Consequently, under linear 0.5/0.5 fusion, lexical exact hits provide the critical discriminatory power that distinguishes relevant candidates from the dense baseline.

Alternatively, Reciprocal Rank Fusion (RRF) is supported in src/retrieval/hybrid.py:46-53 via **Equation (6)**:

$$S_{\mathrm{RRF}}(d) = \frac{w_{\mathrm{BM25}}}{k_{\mathrm{rrf}} + \mathrm{rank}_{\mathrm{BM25}}(d)} + \frac{w_{\mathrm{dense}}}{k_{\mathrm{rrf}} + \mathrm{rank}_{\mathrm{dense}}(d)} \tag{6}$$

Equation (6): S_RRF(d) = [w_BM25 / (k_rrf + rank_BM25(d))] + [w_dense / (k_rrf + rank_dense(d))]
Where: Rank smoothing parameter k_{rrf} = 60, ranks are 1-based. Implemented in src/retrieval/hybrid.py:46-53.

7.2 Cross-Encoder Reranking

Unlike bi-encoders which compute separate vectors, the cross-encoder performs full all-to-all cross-attention across concatenated [claim, passage] tokens, producing an unconstrained scalar logit s_{raw} mapped to relevance via **Equation (7)**:

$$S_{\mathrm{rel}}(c, p) = \sigma(s_{\mathrm{raw}}) = \frac{1}{1 + e^{-s_{\mathrm{raw}}}} \tag{7}$$

Equation (7): S_rel(c, p) = sigma(s_raw) = 1.0 / [1.0 + exp(-s_raw)]
Where: s_{raw} is the single-logit output from cross-encoder/ms-marco-MiniLM-L-6-v2; σ is standard logistic sigmoid mapping $\mathbf{R} \to (0, 1)$. Implemented in src/reranking/cross_encoder.py:56.

7.3 Transformer Natural Language Inference (NLI)

For each surviving candidate passage, local sequence classification model typeform/distilbert-base-uncased-mnli evaluates premise (passage) and hypothesis (claim) via **Equation (8)**:

$$P(y_k \mid p, c) = \frac{\exp(z_k)}{\sum_{j=0}^{2} \exp(z_j)}, \quad k \in \{\mathrm{SUPPORT}, \mathrm{CONTRADICT}, \mathrm{NEUTRAL}\} \tag{8}$$

Equation (8): P(y_k | p, c) = exp(z_k) / [sum_{j=0}^2 exp(z_j)], k in {SUPPORT, CONTRADICT, NEUTRAL}
Where: Raw classification logits $\mathbf{z} \in \mathbf{R}^3$. Model output indices (0: CONTRADICTION, 1: NEUTRAL, 2: ENTAILMENT) are mapped to canonical NLILabel enum and normalized to sum to 1.0. Implemented in src/nli/classifier.py:121-137.

7.4 Evidence Weighting Formulation

In src/aggregation/aggregator.py (lines 62-68), each passage weight is modulated by relevance, temporal validity, and temperature T = 1.0 according to **Equation (9)**:

$$w_p = \left(S_{\mathrm{rel}}(p) \cdot V_{\mathrm{temp}}(p)\right)^{\frac{1}{T}} \tag{9}$$

Equation (9): w_p = (S_rel(p) * V_temp(p))^(1.0 / T)
Where: Relevance score $S_{\text{rel}}(p)$ and temporal validity $V_{\text{temp}}(p)$ are clamped to a minimum floor of 0.01; default temperature T = 1.0. Implemented in src/aggregation/aggregator.py:64.

7.5 Hierarchical Evidence Aggregation

To prevent multiple sentences retrieved from the same document from artificially inflating evidence mass, the aggregator applies intra-document max-pooling across sentences belonging to the same parent document D in **Equations (10)-(12)**:

$$W_D = \max_{p \in \mathcal{P}_D} w_p \tag{10}$$

Equation (10): W_D = max_{p in P_D} w_p
Where: Parent document weight W_D represents peak passage relevance from source document *D*. Implemented in src/aggregation/aggregator.py:96.

$$P_{D,\text{sup}} = \max_{p \in \mathcal{P}_D} P(\text{SUPPORT} \mid p), \quad P_{D,\text{con}} = \max_{p \in \mathcal{P}_D} P(\text{CONTRADICT} \mid p) \tag{11}$$

Equation (11): P_{D,sup} = max_{p in P_D} P(SUPPORT | p), P_{D,con} = max_{p in P_D} P(CONTRADICT | p)
Where: Peak support and contradiction probabilities across all constituent passages P_D from document *D*. Implemented in src/aggregation/aggregator.py:99-106.

$$P_{D,\text{neu}} = \max\left(0.0,\ 1.0 - \left(P_{D,\text{sup}} + P_{D,\text{con}}\right)\right) \tag{12}$$

Equation (12): P_{D,neu} = max(0.0, 1.0 - (P_{D,sup} + P_{D,con}))
Where: Residual neutral probability. Document probabilities are subsequently normalized: $P_{D,c} \leftarrow P_{D,c} / (P_{D,\text{sup}} + P_{D,\text{con}} + P_{D,\text{neu}})$. Implemented in src/aggregation/aggregator.py:108-113.

Independent Pooling Limitation: Because $P_{D,\text{sup}}$ and $P_{D,\text{con}}$ are max-pooled independently, if a document contains a strongly supporting sentence (e.g. 0.9980) alongside a neutral sentence with minor contradiction noise (e.g. 0.0073), their sum exceeds 1.0 (1.0053), forcing renormalization and reducing $P_{D,\text{neu}}$ to zero.

Consensus probabilities across parent documents are synthesized via **Equations (13) and (14)**:

$$P_{\text{agg}}(C) = \frac{\sum_{D \in \mathcal{D}_{\text{pool}}} W_D \cdot P_D(C)}{\sum_{D \in \mathcal{D}_{\text{pool}}} W_D}, \quad C \in \{\text{SUP}, \text{CON}, \text{NEU}\} \tag{13}$$

Equation (13): P_agg(C) = [sum_{D in D_pool} W_D * P_D(C)] / [sum_{D in D_pool} W_D], C in {SUP, CON, NEU}
Where: Document-weighted class probability distribution across all represented parent sources D_{pool}. Implemented in src/aggregation/aggregator.py:143-151.

$$\text{Effective_Mass} = \sum_{D \in \mathcal{D}_{\text{pool}}} W_D \tag{14}$$

Equation (14): Effective_Mass = sum_{D in D_pool} W_D
Where: Total accumulated evidence mass across distinct sources. If Effective_Mass ≤ 0.0, decider enforces Rule 1 abstention. Implemented in src/aggregation/aggregator.py:118, 169.

Corroboration Invariance: Because Equation (13) normalizes by $\sum W_D$, ten agreeing documents produce the exact same consensus probability distribution as a single document with the same entailment score. Multi-document corroboration increases effective mass but does not increase consensus probability.

7.6 Normalized Conflict-Balance Score

In src/conflict/detector.py, evidence passages are deduplicated by text. Supporting and contradicting masses are accumulated over unique passages categorized by predominant argmax NLI label in **Equation (15)**:

$$M_{\text{sup}} = \sum_{p \in \mathcal{P}_{\text{sup}}} S_{\text{rel}}(p) \cdot P(\text{SUP} \mid p), \quad M_{\text{con}} = \sum_{p \in \mathcal{P}_{\text{con}}} S_{\text{rel}}(p) \cdot P(\text{CON} \mid p) \tag{15}$$

Equation (15): M_sup = sum_{p in P_sup} S_rel(p) * P(SUP | p), M_con = sum_{p in P_con} S_rel(p) * P(CON | p)
Where: P_{sup} contains passages where argmax label is SUPPORT; P_{con} contains passages where argmax label is CONTRADICT. Note: temporal validity V_{temp} is not included in conflict mass accumulation. Implemented in src/conflict/detector.py:57-73.

The normalized conflict-balance score is formulated in **Equation (16)**:

$$\text{Conflict_Score} = \frac{2 \cdot \min(M_{\text{sup}},\, M_{\text{con}})}{M_{\text{sup}} + M_{\text{con}}} \tag{16}$$

Equation (16): Conflict_Score = [2.0 * min(M_sup, M_con)] / [M_sup + M_con]
Where: Normalized conflict-balance ratio in [0.0, 1.0]. Represents parity between opposing evidence masses: perfectly balanced opposing masses yield 1.00; single-sided consensus yields 0.00. Scale-invariant: two weak opposing masses of 0.05 produce a score of 1.00. Zero-denominator behavior: returns 0.0 if total mass is 0. Implemented in src/conflict/detector.py:79-83.

The binary conflict condition is triggered via **Equation (17)**:

$$\text{conflict_detected} = (\text{Conflict_Score} \geq \tau_{\text{bal}}) \wedge (|\mathcal{P}_{\text{sup}}| > 0) \wedge (|\mathcal{P}_{\text{con}}| > 0) \tag{17}$$

Equation (17): conflict_detected = (Conflict_Score >= tau_bal) and (|P_sup| > 0) and (|P_con| > 0)
Where: Threshold Distinction: The Python class signature default in src/conflict/detector.py:24 is balance_threshold = 0.25 (active when ConflictDetector() is instantiated without arguments in standalone unit tests). In active pipeline execution, configs/default.yaml:45 and configs/controlled.yaml:45 configure balance_threshold: 0.30 (passed via src/pipeline/verifier.py:108). The operational verification threshold is therefore $\tau_{\text{bal}} = 0.30$. Requires at least one supporting and one contradicting passage ($|P_{\text{sup}}| \geq 1$ and $|P_{\text{con}}| \geq 1$). Note: 0.60 belongs strictly to Rule 3's uncertainty check. Implemented in src/conflict/detector.py:85-89.

7.7 Chronological Analysis and Temporal Validity Decay

In src/temporal/analyzer.py, the analyzer extracts 4-digit years from claim and evidence text via regex. When a claim specifies a target year Y_{claim} and evidence specifies different event years without mentioning Y_{claim}, the timeline gap Δt triggers linear decay via **Equation (18)**:

$$\text{mismatch}}} \cdot \min\left(1.0,\, \frac{\Delta t}{10.0}\right)\right), \quad \Delta t = \min_{y \in \mathrm{mathca} \tag{18}$$

Equation (18): V_temp(p) = max(0.3, 1.0 - mu_mismatch * min(1.0, Delta_t / 10.0)), Delta_t = min_{y in Y_text} |Y_claim - y|
Where: Rule 1 -- Year-Gap Linear Decay: Applies when a claim explicitly specifies a calendar year Y_{claim}. For evidence event years Y_{text} not containing Y_{claim}, timeline gap $\Delta t = \min |Y_{\text{claim}} - y| \geq 2$ years triggers decay. Inactive Floor Explanation: With active config $\mu_{\text{mismatch}} = 0.50$, max penalty is $0.50 \cdot 1.0 = 0.50$; thus $1.0 - 0.50 = 0.50 > 0.30$. The value inside max cannot fall below 0.50; hence, the 0.30 floor clamp is strictly inactive and values span [0.50, 1.00]. If any passage in the retrieved pool has $\Delta t \geq 2$, has_mismatch = True and $\text{Penalty}_{\text{temp}} = 0.5000$. Implemented in src/temporal/analyzer.py:91-110.

For present-oriented claims (containing markers such as 'currently', 'today', 'market leader') where older evidence precedes newer evidence by ≥ 3 years, validity decays according to **Equation (19)**:

$$\text{}}}(p) = \max(0.2,\, 1.0 - \mu_{\mathrm{outdated}}) = 1.0 - 0.30 = 0.70 \quad (\Delta t_{\mathrm{pub}} \tag{19}$$

Equation (19): V_temp(p) = max(0.2, 1.0 - mu_outdated) = 1.0 - 0.30 = 0.70
Where: Rule 2 -- Present-Oriented Contemporary Rule: Applies when is_present is True (claim contains contemporary markers like 'currently', 'today', 'market leader' and no explicit years), and older evidence is superseded by newer evidence in the pool by ≥ 3 years (max $Y_{\text{ev}} - Y_{\text{pub}} \geq 3$). Validity decays to $V_{\text{temp}} = \max(0.2, 1.0 - 0.30) = 0.70$ (0.20 floor inactive). If triggered, has_mismatch = True and pool penalty is $\text{Penalty}_{\text{temp}} = \mu_{\text{outdated}} = 0.3000$. Implemented in src/temporal/analyzer.py:84-90, 110.

7.8 Multi-Signal Epistemic Uncertainty Estimation

Uncertainty is synthesized from four distinct epistemic shortfall signals in src/uncertainty/estimator.py:

1. **Normalized Shannon Entropy:** Measures dispersion across the 3-class aggregate NLI probability vector according to **Equation (20)**:

$$u_{\text{entropy}} = -\frac{1}{\ln(3.0)} \sum_{k \in \{\text{SUP, CON, NEU}\}} p_k \cdot \ln(p_k + 10^{-12}) \quad \textbf{(20)}$$

Equation (20): u_entropy = - [sum_{k in {SUP, CON, NEU}} p_k * ln(p_k + 10^-12)] / ln(3.0)
Where: $p_k = P_{\text{agg}}(k)$; division by ln(3.0) ≈ 1.0986 normalizes Shannon entropy into [0.0, 1.0]. Note: if all retrieved evidence evaluates to 100% NEUTRAL, entropy is 0.00 (near-zero uncertainty), and the verdict falls through to Rule 6 default abstention. Implemented in src/uncertainty/estimator.py:50-54.

2. **Conflict Uncertainty:** u_{conflict} = clip(Conflict_Score, 0.0, 1.0).
3. **Retrieval Deficit Uncertainty:** Measures evidence weakness relative to required count $N_{\text{min}} = 1$ according to **Equation (21)**:

rm{eff}} \le 0 \\ \max\left(0.0,\, 1.0 - \max_{p \in \mathcal{P}} S_{\mathrm{rel}}(p) \cdot \min\left(1.0,\ **(21)**

Equation (21): u_retrieval = 1.0 (if |P|=0), max(0.0, 1.0 - max S_rel * min(1.0, |P| / N_min)) (otherwise)
Where: Early Exit: If $|P| = 0$ or Effective_Mass ≤ 0, the system immediately returns $u_{\text{retrieval}} = 1.0$, Total_U = 1.0, and Calibrated_Confidence = 0.0 without invoking downstream NLI (src/uncertainty/estimator.py:64-78). General Formula: $u_{\text{retrieval}} = \max(0.0, \min(1.0, 1.0 - (\max S_{\text{rel}} \cdot \min(1.0, |P| / N_{\text{min}}))))$. Because $N_{\text{min}} = \max(1, \text{min_evidence_count}) = 1$ by default, for any non-empty evidence set ($|P| \geq 1$), the count factor min(1.0, |P| / 1) is identically 1.0. Thus, retrieval deficit simplifies directly to $u_{\text{retrieval}} = 1.0 - \max_{p \in P} S_{\text{rel}}(p)$. Safeguard Floor (Equation 23): Total_U ≥ 0.65 · $u_{\text{retrieval}}$ ensures that missing evidence ($u_{\text{retrieval}} = 1.0$) guarantees Total_U ≥ 0.65, exceeding $\tau_{\text{abstain}} = 0.60$ and decisively forcing Rule 3 abstention. Implemented in src/uncertainty/estimator.py:91-106.

4. **Temporal Penalty:** u_{temporal} = clip(Penalty$_{\text{temp}}$, 0.0, 1.0).
Composite Uncertainty: In lines 99-106, signals are combined using normalized weights $w_{\text{ent}} = 0.35$, $w_{\text{conf}} = 0.35$, $w_{\text{retr}} = 0.20$, $w_{\text{temp}} = 0.10$ via **Equation (22)**, subject to the retrieval deficit safeguard floor formulated in **Equation (23)**:

$$u_{\text{weighted}} = 0.35 \cdot u_{\text{entropy}} + 0.35 \cdot u_{\text{conflict}} + 0.20 \cdot u_{\text{retrieval}} + 0.10 \cdot u_{\text{temporal}} \quad \textbf{(22)}$$

Equation (22): u_weighted = 0.35*u_entropy + 0.35*u_conflict + 0.20*u_retrieval + 0.10*u_temporal
Where: Heuristic weighted sum of entropy, conflict, retrieval deficit, and temporal penalty weights (sum = 1.0). Implemented in src/uncertainty/estimator.py:99-104.

$$\text{Total_Uncertainty} = \text{clip}(\max(u_{\text{weighted}},\ 0.65 \cdot u_{\text{retrieval}}),\ 0.0,\ 1.0) \quad \textbf{(23)}$$

Equation (23): Total_Uncertainty = clip(max(u_weighted, 0.65 * u_retrieval), 0.0, 1.0)
Where: Safeguard floor (0.65 · $u_{\text{retrieval}}$) guarantees that ungrounded claims lacking evidence cannot have low uncertainty. Implemented in src/uncertainty/estimator.py:105-106.

Operational Confidence Score: In line 107, operational confidence is defined as the complement of total uncertainty according to **Equation (24)**:

$$\text{Confidence} = \text{round}(\text{clip}(1.0 - \text{Total_Uncertainty},\ 0.0,\ 1.0),\ 4) \quad \textbf{(24)}$$

Equation (24): Confidence = round(clip(1.0 - Total_Uncertainty, 0.0, 1.0), 4)
Where: Operational decision score in [0.0, 1.0] indicating absence of measured uncertainty signals; not an empirically calibrated Bayesian probability of truth. Implemented in src/uncertainty/estimator.py:107.

SCIENTIFIC SAFEGUARD: OPERATIONAL SCORE VS. CALIBRATED PROBABILITY

RIGOROUS SCIENTIFIC DISTINCTION: Confidence is an *operational decision score*, NOT a calibrated Bayesian probability that the claim is factually true. It reflects the absence of measured uncertainty signals (dispersion, conflict, retrieval deficit, timeline gap). It must not be confused with (a) NLI softmax class probability *P*(SUP), (b) cumulative evidence mass, or (c) an empirically calibrated posterior fitted via Platt scaling or isotonic regression.

7.9 Decision Rules and Abstention Policy

In src/decision/decider.py (lines 38-135), the VerificationDecider evaluates a strict hierarchy of deterministic rules. Abstention rules precede decisive classification, ensuring that ungrounded, contradictory, or ambiguous inputs trigger *INSUFFICIENT_EVIDENCE*. Polar evidence mass is defined as $P_{polar} = P_{sup} + P_{con}$.

Rule & Priority	Exact Evaluated Condition	Resulting Verdict	Source Code Reference
Rule 1: Zero Evidence Mass	Effective_Mass <= 0.0 (or no passages retrieved)	INSUFFICIENT_EVIDENCE	src/decision/decider.py:39
Rule 2: High Conflict Trigger	enable_abstention == True and conflict.conflict_detected == True (Fires when Conflict_Score >= 0.30 with opposing evidence pools)	INSUFFICIENT_EVIDENCE	src/decision/decider.py:51
Rule 3: High Uncertainty Trigger	enable_abstention == True and Total_Uncertainty > max_uncertainty_threshold (0.60)	INSUFFICIENT_EVIDENCE	src/decision/decider.py:62
Rule 4: Decisive SUPPORTED	(P_sup >= 0.55 and P_sup > P_con) OR (P_polar >= 0.25 and P_sup/P_polar >= 0.70 and P_sup > 2*P_con and not conflict)	SUPPORTED	src/decision/decider.py:73-85
Rule 5: Decisive CONTRADICTED	(P_con >= 0.55 and P_con > P_sup) OR (P_polar >= 0.25 and P_con/P_polar >= 0.70 and P_con > 2*P_sup and not conflict)	CONTRADICTED	src/decision/decider.py:95-106
Rule 6: Fallback Default	All conditions above fail; evidence does not meet decisive thresholds	INSUFFICIENT_EVIDENCE	src/decision/decider.py:128

Ratio-Based Branch Analysis: The second branch of Rules 4 and 5 ($P_{polar} \geq 0.25$ and ratio ≥ 0.70) is more aggressive than the direct 55% threshold. For example, if $P_{sup} = 0.18$, $P_{con} = 0.07$, and $P_{neu} = 0.75$, then $P_{polar} = 0.25$, ratio = 0.18/0.25 = 0.72 $\geq$ 0.70, and $P_{sup} > 2 \cdot P_{con}$ (0.18 > 0.14). This branch fires SUPPORTED even when 75% of the evidence mass is neutral, preventing weak neutral filler from suppressing decisive polar evidence.

7.10 Evaluation Metrics and Their Formulas

All benchmark metrics reported across experiments are computed in src/evaluation/metrics.py:

• **Multi-Class Classification Accuracy:** Fraction of total test instances where predicted verdict matches ground truth, formulated in **Equation (25)**:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(y_i = \hat{y}_i) \tag{25}$$

Equation (25): Accuracy = [sum_{i=1}^N I(y_i == y^_i)] / N
Where: N is total test claims; y_i, $y^_i \in$ {SUPPORTED, CONTRADICTED, INSUFFICIENT_EVIDENCE}; $\mathbf{1}(\cdot)$ is the indicator function. Implemented in src/evaluation/metrics.py:136.

• **Macro-Averaged F1 Score:** Unweighted arithmetic mean of per-class F1 harmonic means across the 3 canonical classes according to **Equation (26)**:

$$\text{Macro_F1} = \frac{1}{3} \sum_{c \in \{\text{SUP}, \text{CON}, \text{INSUF}\}} \text{F1}_c, \quad \text{F1}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{26}$$

Equation (26): Macro_F1 = (1 / 3) * [F1_SUPPORTED + F1_CONTRADICTED + F1_INSUFFICIENT]
Where: Evaluates balanced multi-class performance giving equal weight to each verification outcome regardless of class prevalence. Implemented in src/evaluation/metrics.py:150.

• **Prediction Decision Coverage:** Proportion of instances where the system commits to a definitive verdict rather than abstaining, defined in **Equation (27)**:

$$\text{Coverage} = \frac{|\mathcal{C}_{\text{covered}}|}{N} = \frac{|\{i : \hat{y}_i \neq \text{INSUFFICIENT_EVIDENCE}\}|}{N} \tag{27}$$

Equation (27): Coverage = |C_covered| / N = |{i : y^_i != INSUFFICIENT_EVIDENCE}| / N
Where: Measures the empirical willingness of the model to issue definitive claims under confidence and conflict constraints. Implemented in src/evaluation/metrics.py:239.

• **Selective Accuracy (Conditional Accuracy on Accepted Predictions):** Evaluated strictly over non-abstained instances according to **Equation (28)**:

$$\text{Selective_Accuracy} = \frac{1}{|\mathcal{C}_{\text{covered}}|} \sum_{i \in \mathcal{C}_{\text{covered}}} \mathbb{I}(y_i = \hat{y}_i) \quad (|\mathcal{C}_{\text{covered}}| > 0;\ \text{undefined if } 0) \tag{28}$$

Equation (28): Selective_Accuracy = [sum_{i in C_covered} I(y_i == y^_i)] / |C_covered| (|C_covered| > 0; Undefined if 0)
Where: Computes *P*(correct | accepted). If $|C_{covered}| = 0$ (as in Exp A baseline), selective accuracy is mathematically undefined (0/0) and programmatically stored as 0.0 fallback in src/evaluation/metrics.py:245. Distinct from precision: selective accuracy is the conditional accuracy over covered claims. Implemented in src/evaluation/metrics.py:240-247.

• **Retrieval Recall@k and Mean Reciprocal Rank (MRR):** Evaluated against alternative gold evidence sets in FEVER according to **Equations (29) and (30)**:

$$\text{Recall@}k = \frac{1}{N_{\text{gold}}} \sum_{i=1}^{N_{\text{gold}}} \max_{\mathcal{G} \in \text{GoldSets}_i} \frac{|\text{Top}_k(i) \cap \mathcal{G}|}{|\mathcal{G}|} \tag{29}$$

Equation (29): Recall@k = (1/N_gold) * sum_i max_{G in GoldSets_i} [|Top_k_i cap G| / |G|]
Where: GoldSets_i contains alternative valid evidence sets in FEVER; evaluated over top-*k* retrieved candidate passages (*k* = 5). Implemented in src/evaluation/metrics.py:176-189.

$$\text{MRR} = \frac{1}{N_{\text{gold}}} \sum_{i=1}^{N_{\text{gold}}} \frac{1}{\text{rank}_{\text{first_gold}}(i)} \tag{30}$$

Equation (30): MRR = (1/N_gold) * sum_i [1.0 / first_gold_rank_i]
Where: $\text{rank}_{\text{first_gold}}(i)$ is 1-based rank of first relevant gold passage; score is 0.0 if no gold passage appears in top-20. *Identity Explanation:* In FEVER evaluation, Recall@5 and MRR are both exactly 0.225 because for all 9 claims where gold evidence was retrieved in the top 5, the first gold document appeared at rank 1 (1/1.0 = 1.0), while for the remaining 31 verifiable claims no gold passage was retrieved in top-20 (0.0), producing identical means: 9/40 = 0.225. Implemented in src/evaluation/metrics.py:192-205.

• **Expected Calibration Error (ECE):** Evaluated across $B = 5$ confidence bins according to **Equation (31)**:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \cdot |\text{acc}(I_b) - \text{conf}(I_b)| \tag{31}$$

Equation (31): ECE = sum_{b=1}^B (|I_b| / N) * | acc(I_b) - conf(I_b) |
Where: Predictions partitioned into $B = 5$ confidence bins $I_b = ((b\text{-}1)/B, b/B]$; acc($I_b$) is empirical accuracy; conf(I_b) is mean predicted confidence in bin *b*. Implemented in src/evaluation/metrics.py:214-228.

• **Multiclass Brier Score:** Mean squared error between predicted probability vectors and one-hot true vectors, defined in **Equation (32)**:

$$\text{BS}_{\text{raw}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{3} (p_{i,k} - y_{i,k})^2 \in [0, 2], \quad \text{BS}_{\text{norm}} = \frac{\text{BS}_{\text{raw}}}{2.0} = \frac{1}{2N} \sum_{i=1}^{N} \sum_{k=1}^{3} (p_{i,k} - y_{i,k})^2 \in [0, 1] \tag{32}$$

Equation (32): Brier = (1 / N) * sum_{i=1}^N sum_{k=1}^3 (p_{i,k} - y_{i,k})^2, Normalized = Brier / 2.0
Where: Reconciliation of /2N vs /N: In academic literature, the multiclass Brier score is often written with a 1/(2N) divisor to normalize the metric into [0.0, 1.0]. In src/evaluation/metrics.py:196, the un-normalized sum of squared Euclidean deviations is divided by N, yielding Raw Multiclass Brier: $BS_{raw} = (1/N) \sum_{i=1}^{N} \sum_{k=1}^{3} (p_{i,k} - y_{i,k})^2 \in [0.0, 2.0]$ (reported as 1.2544 on FEVER and 0.7115 on Controlled). Dividing BS_{raw} by 2.0 (metrics.py:198) yields Normalized Brier: $BS_{norm} = BS_{raw} / 2.0 = (1/2N) \sum \sum (p - y)^2 \in [0.0, 1.0]$ (reported as 0.6272 on FEVER and 0.3558 on Controlled). Summary tables report both values side-by-side under 'Raw Brier / Norm'. Implemented in src/evaluation/metrics.py:182-215.

7.11 Additional Mathematical Operations in Code

• **Subclaim Conjunction Rule:** In src/pipeline/verifier.py:340-355, compound claims decomposed into subclaims $\{s_1, ..., s_m\}$ follow strict logical conjunction: the compound claim is *SUPPORTED* if and only if all subclaims are *SUPPORTED*; if any subclaim is *CONTRADICTED*, the claim is *CONTRADICTED*; otherwise, it abstains with *INSUFFICIENT_EVIDENCE*.
• **Compound Claim Average Confidence and Uncertainty:** In src/pipeline/verifier.py:356-357, compound confidence is the arithmetic mean across constitutive subclaim confidences: $\text{Confidence}_{comp} = (1/m) \sum_j \text{Confidence}(s_j)$, and compound uncertainty is its complement: $\text{Uncertainty}_{comp} = 1.0 - \text{Confidence}_{comp}$.
• **Cosine Conflict Angle:** When evaluating pairwise passage divergence in test fixtures, divergence angle is computed as $\theta = \arccos(\mathbf{v}_1 \cdot \mathbf{v}_2 / (\|\mathbf{v}_1\|_2 \|\mathbf{v}_2\|_2))$.

7.12 Formula-to-Code Traceability Table

The following matrix maps every mathematical equation directly to its implementing Python module, configuration parameter, and empirical effect on the verdict:

Concept	Formula Ref	Variables & Domain	Implementing File & Line	Config Parameter	Effect on Final Verdict	Status
Dense Similarity	Equation (1-2)	q, d in R^384; S in [0, 1]	src/retrieval/dense.py:103-111	dense_model_name	Selects initial candidate documents	Verified
BM25 Saturation	Equation (3-4)	k1=1.5, b=0.75, c=10.0	src/retrieval/bm25.py:72-84	corpus_path	Extracts exact keyword matches	Verified
Linear Fusion	Equation (5)	w_bm25=0.5, w_dense=0.5	src/retrieval/hybrid.py:29-80	hybrid_weight_bm25	Balances keyword and semantic hits	Verified
Cross-Encoder	Equation (7)	s_raw in R; S_rel in [0, 1]	src/reranking/cross_encoder.py:56	reranking.top_k: 5	Ranks top-5 passages for NLI	Verified
NLI Softmax	Equation (8)	z in R^3; P in [0, 1], sum=1	src/nli/classifier.py:121-137	nli.model_name	Yields passage premise entailment	Verified
Passage Weight	Equation (9)	S_rel, V_temp in [0, 1], T=1.0	src/aggregation/aggregator.py:64	relevance_temperature	Modulates mass by relevance & time	Verified
Doc-Aware Max	Equation (10-12)	W_D = max w_p; P_D in [0, 1]	src/aggregation/aggregator.py:96-108	pooling_mode: doc_aware	Prevents multi-passage mass inflation	Verified
Consensus Prob	Equation (13-14)	W_D in [0, 1]; P_agg in [0, 1]	src/aggregation/aggregator.py:143-151	aggregation.method	Produces aggregate evidence signal	Verified
Conflict Score	Equation (15-17)	M_sup, M_con in R+; Score in [0,1]	src/conflict/detector.py:81-88	balance_threshold: 0.30 (default 0.25)	Triggers Rule 2 abstention if score >= 0.30	Verified
Temporal Decay	Equation (18-19)	dt in N; mu=0.5; V_temp in [0.5, 1.0]	src/temporal/analyzer.py:101-106	future_mismatch_penalty: 0.5	Penalizes outdated or future evidence	Verified
Shannon Entropy	Equation (20)	P in [0, 1]^3; u_ent in [0, 1]	src/uncertainty/estimator.py:50-54	entropy_weight: 0.35	Increases uncertainty on flat NLI	Verified
Retrieval Deficit	Equation (21)	S_rel in [0, 1]; u_retr in [0, 1]	src/uncertainty/estimator.py:93	retrieval_deficit_weight	Forces uncertainty on missing evidence	Verified
Composite Uncert	Equation (22-23)	w in R^4; Total_U in [0, 1]	src/uncertainty/estimator.py:99-106	max_uncertainty: 0.60	Triggers Rule 3 abstention if Total_U > 0.60	Verified
Operational Conf	Equation (24)	Conf = 1.0 - Total_U	src/uncertainty/estimator.py:107	N/A (Derived score)	Reported decision metric in UI	Verified
Selective Accuracy	Equation (28)	N_covered in N; Sel_Acc in [0, 1]	src/evaluation/metrics.py:242	abstention_label	Evaluates selective accuracy (conditional accuracy) on accepted claims	Verified

7.13 Worked Numerical Examples with Intermediate Arithmetic

The following four worked examples demonstrate the exact arithmetic execution of the mathematical formulas using actual logged values from verified pipeline runs.

Worked Example 1: Supported Claim (Demo A -- Margaret Thatcher)

• **Input Assertion:** Claim c = *'Margaret Thatcher served as Prime Minister of the United Kingdom.'*
• **Passage #0 Extraction:** Document wiki_margaret_thatcher, sentence 0 (S_{rel} = 1.000, V_{temp} = 1.000).
• **Weight Calculation (Equation 9):** $w_0 = (1.000 \cdot 1.000)^{1/1.0} = 1.000$.
• **Stage 1 -- Local DistilBERT Sentence NLI Softmax (Equation 8):** $P(\text{SUP} \mid s_0) = 0.9980$, $P(\text{CON} \mid s_0) = 0.0003$, $P(\text{NEU} \mid s_0) = 0.0017$.
• **Stage 2 -- Intra-Document Max-Pooling & Renormalization (Equations 10-12):** Across 5 retrieved sentences from the same document (sentences 2, 8, 1, 13 evaluate to NEUTRAL):
$P_{D,\text{sup}}$ = max(0.9980, 0.0282, 0.0090, 0.0016, 0.0007) = 0.9980.
$P_{D,\text{con}}$ = max(0.0003, 0.0073, 0.0054, 0.0064, 0.0011) = 0.0073.
$P_{D,\text{neu}}$ = max(0.0, 1.0 - (0.9980 + 0.0073)) = max(0.0, -0.0053) = 0.0.
Normalization: norm = 0.9980 + 0.0073 = 1.0053 → $P_{D,\text{sup}}$ = 0.9980/1.0053 = **0.9927**, $P_{D,\text{con}}$ = **0.0073**.
(Traceability Note: 0.9980 is the un-normalized sentence-0 NLI output; 0.9927 is the document-level aggregated support probability after max-pooling and renormalization with sentence-1 contradiction mass).
• **Normalized Shannon Entropy (Equation 20):**
H = -[0.9927 · ln(0.9927) + 0.0073 · ln(0.0073) + 0] = -[0.9927 · (-0.0073) + 0.0073 · (-4.9199)] = 0.0431.
Normalized: u_{entropy} = 0.0431 / ln(3.0) = 0.0431 / 1.0986 = **0.0393**.
• **Uncertainty Composition (Equations 21-23):** u_{conflict} = 0.0, $u_{\text{retrieval}}$ = 1.0 - 1.000 = 0.0, u_{temporal} = 0.0.
Total_U = 0.35 · 0.0393 + 0 + 0 + 0 = **0.0138**.
• **Stage 3 -- Operational Decision Confidence (Equation 24):** Confidence = 1.0 - 0.0138 = **0.9862 (98.62%)**.
• **Decision:** Mass = 1.000 > 0; Conflict = 0.00 < 0.30; Total_U = 0.0138 ≤ 0.60; P_{sup} = 0.9927 ≥ 0.55 → **SUPPORTED**.

Worked Example 2: Contradicted Claim (Demo B -- Thatcher Negated)

• **Input Assertion:** Claim c = *'Margaret Thatcher was never Prime Minister of the United Kingdom.'*
• **Passage #0 Extraction:** Same authoritative document (S_{rel} = 0.9998, V_{temp} = 1.000).
• **Stage 1 -- Local DistilBERT Sentence NLI Softmax (Equation 8):** $P(\text{SUP} \mid s_0) = 0.0000$, $P(\text{CON} \mid s_0) = 0.9998$, $P(\text{NEU} \mid s_0) = 0.0001$.
• **Stage 2 -- Intra-Document Max-Pooling & Renormalization (Equations 10-12):** $P_{D,\text{con}}$ = 0.9998, $P_{D,\text{sup}}$ = 0.0001.
• **Normalized Shannon Entropy (Equation 20):** u_{entropy} = **0.0019**.
• **Uncertainty Composition (Equations 21-23):** $u_{\text{retrieval}}$ = 1.0 - 0.9998 = 0.0002; u_{conflict} = 0.0; u_{temporal} = 0.0.
Total_U = 0.35 · 0.0019 + 0.20 · 0.0002 = 0.00066 + 0.00004 = **0.0007**.
• **Operational Decision Score (Equation 24):** Confidence = 1.0 - 0.0007 = **0.9993 (99.93%)**.
• **Decision:** Mass = 0.9998 > 0; Conflict = 0.00; Total_U = 0.0007 ≤ 0.60; P_{con} = 0.9998 ≥ 0.55 → **CONTRADICTED**.

Worked Example 3: Conflicting Controlled Scenario (Demo C -- NovaTech Revenue)

• **Input Assertion:** Claim *c* = *'NovaTech enterprise platform generated $10 million in undisputed 2024 earnings.'*
• **Supporting Passage:** Press release (S_{rel} = 0.9974, P_{sup} = 0.9974, P_{con} = 0.0005) → M_{sup} = 0.9974.
• **Contradicting Passage:** Audit report (S_{rel} = 0.9964, P_{sup} = 0.0023, P_{con} = 0.9964) → M_{con} = 0.9964.
• **Normalized Conflict-Balance Score (Equation 16):**
Conflict_Score = [2.0 · min(0.9974, 0.9964)] / [0.9974 + 0.9964] = [2.0 · 0.9964] / [1.9938] = 1.9928 / 1.9938 = **0.9995 (≈ 1.00; reported 0.98 across candidate pool)**.
(Reconciliation Note: In the static benchmark results/controlled/full_system.json, candidate pool aggregation yielded 50.8% support vs 47.5% contradict, logging Conflict_Score = 0.98, Confidence = 36.43%, and Total_U = 63.57%. In isolated top-2 pairwise calculation, harmonic balance gives 0.9995 ≈ 1.00, Confidence = 37.56%, Total_U = 62.44%; in live CLI execution, Conflict_Score = 1.00 [0.9989] and Confidence = 37.50%. Both trigger identical Rule 2 abstention).
• **Conflict Trigger (Equation 17):** Conflict_Score (0.98 / 1.00) ≥ τ_{bal} (0.30) and $|P_{sup}| \geq 1$ and $|P_{con}| \geq 1$ → **conflict_detected = True**.
• **Candidate Pool Chronological Penalty:** While the two opposing NovaTech documents are temporally aligned to 2024, retrieved candidate passages #4 and #5 reference 2019 and 2021. Because TemporalAnalyzer assesses the entire retrieved candidate pool, passages #4 and #5 (Acme Corp 2019 and Mars rover 2021) have $\Delta t \geq 2$ relative to 2024, triggering has_mismatch = True in TemporalAnalyzer:103 → $Penalty_{temp}$ = $\mu_{mismatch}$ = 0.5000 and $u_{temporal}$ = **0.5000**. In an isolated 2-doc evaluation (looking only at the press release and audit report), $\Delta t = 0$ → $u_{temporal}$ = 0.0000. In both cases, Rule 2 abstention triggers decisively on conflict detection regardless of $u_{temporal}$.
• **Uncertainty Composition (Equations 20-23):** $u_{entropy}$ = 0.6413, $u_{conflict}$ = 0.9995, $u_{retrieval}$ = 0.0007, $u_{temporal}$ = 0.5000.
Total_U = 0.35 · 0.6413 + 0.35 · 0.9995 + 0.20 · 0.0007 + 0.10 · 0.5000 = 0.2245 + 0.3498 + 0.0001 + 0.0500 = **0.6244 (62.44%)**.
• **Operational Decision Score (Equation 24):** Confidence = 1.0 - 0.6244 = **0.3756 (37.56%)**.
• **Decision:** Rule 2 triggers immediately because conflict_detected == True (Conflict_Score = 0.98 ≥ 0.30) → **INSUFFICIENT_EVIDENCE**.
(Even if Rule 2 were bypassed, Rule 3 would independently trigger because Total_U = 0.6244 > 0.60 threshold).

Worked Example 4: Controlled Diagnostic Suite Metrics Calculation (N=9 Claims)

• **Total Evaluated Claims:** *N* = 9 (3 Supported, 4 Contradicted, 2 Insufficient Evidence).
• **Full System (Exp D) Predictions Breakdown:**
- Definitive verdicts issued (Covered): 4 claims (James Webb → SUP [correct], Nobel Chemistry → SUP [correct], Acme Drones → CON [correct], Alan Turing → CON [correct]).
- Abstained verdicts issued: 5 claims:
1. NovaTech → INSUF (correct abstention due to 0.98 conflict score triggering Rule 2).
2. Mars 1000 BC → INSUF (correct abstention due to 100% neutral evidence triggering Rule 6).
3. Roman Atwood → INSUF (error of omission: decomposed into 2 subclaims; subclaim verifier lacked decisive evidence for second subclaim, triggering conjunction abstention).
4. Eiffel Tower → INSUF (error of omission: retrieved evidence mass was 80.2% NEUTRAL, P_{con} = 13.2% < 55% threshold, P_{polar} = 0.198 < 0.25, falling through to Rule 6 default abstention).
5. Perseverance → INSUF (error of omission: evidence was 70.0% NEUTRAL, P_{con} = 20.0%, failing ratio threshold 0.20/0.30 = 0.667 < 0.70, falling through to Rule 6 default abstention).
• **Coverage Calculation (Equation 27):** Coverage = $|C_{covered}|$ / *N* = 4 / 9 = **0.4444 (44.44%)** (exact 95% CI: [13.7%, 78.8%]).
• **Selective Accuracy Calculation (Equation 28):** Across the 4 covered instances, all 4 match ground truth:
Selective_Accuracy = 4 / 4 = **1.0000 (100.00%)** (exact 95% Clopper-Pearson CI: [39.8%, 100.0%]).
• **3-Class Raw Accuracy (Equation 25):** Total correct = 2 (SUP) + 2 (CON) + 2 (INSUF) = 6 correct out of 9:
Accuracy = 6 / 9 = **0.6667 (66.67%)** (exact 95% CI: [29.9%, 92.5%]).
• **Evidence-Aware Comparison (Exp B):** Predicts on all 9 claims without abstention (100% coverage). Correct on all 7 verifiable claims (3 Supported, 4 Contradicted), but falsely predicts SUPPORTED on both ungrounded claims (NovaTech and Mars 1000 BC), achieving 7/9 = 77.78% accuracy.
• **Baseline Comparison (Exp A):** Predicts INSUFFICIENT for all 9 claims (always-abstain baseline). $|C_{covered}|$ = 0.
Coverage = 0 / 9 = 0.0%. Selective_Accuracy = 0 / 0 = **Undefined** (programmatically stored as 0.0 fallback in metrics.py:245).

8. Demonstrations and Reproducibility

To establish reproducible evidence, three canonical demonstrations were executed directly against the local repository using the live CLI tool scripts/run_pipeline.py.

Demo A: Supported Claim (Demonstrated Historical Case Study)

• **Execution Command:** python scripts/run_pipeline.py "Margaret Thatcher served as Prime Minister of the United Kingdom."

• **Execution Mode:** Reproduced full-pipeline result (evaluated on the default FEVER corpus).

• **Final Verdict:** SUPPORTED | **Operational Decision Score:** 98.62% | **Document-Level Aggregate Support:** 99.27% | **Sentence NLI Support:** 99.80% | **Total Uncertainty:** 1.38% | **Conflict:** False (0.00)

• **Retrieved Evidence Span:** Document wiki_margaret_thatcher, sentence 0 (Published: 2021-01-01):
'Margaret Hilda Thatcher, Baroness Thatcher (nee Roberts; 13 October 1925 - 8 April 2013) was a British stateswoman who served as Prime Minister of the United Kingdom from 1979 to 1990 and Leader of the Conservative Party from 1975 to 1990.'

• **Local DistilBERT NLI Output:** P(SUPPORT) = 0.9980, P(CONTRADICT) = 0.0003, P(NEUTRAL) = 0.0017.

• **Architectural Note:** Four additional sentences retrieved from the same document evaluated to NEUTRAL (P_neu >= 0.96). Document-aware max-pooling correctly isolated the decisive premise without dilution. Note that multiple sentences from this document are not independent sources; they represent a single corroborated source document.

Demo B: Negated Claim (Explicit Refutation Premise)

• **Execution Command:** python scripts/run_pipeline.py "Margaret Thatcher was never Prime Minister of the United Kingdom."

• **Execution Mode:** Reproduced full-pipeline result (evaluated on the default FEVER corpus).

• **Final Verdict:** CONTRADICTED | **Operational Score:** 99.93% | **Total Uncertainty:** 0.07% | **Conflict:** False (0.00)

• **Retrieved Evidence Span:** Same authoritative document (wiki_margaret_thatcher#s0).

• **Local DistilBERT NLI Output:** P(SUPPORT) = 0.0000, P(CONTRADICT) = 0.9998, P(NEUTRAL) = 0.0001.

• **Architectural Note:** Demonstrates premise-hypothesis asymmetry: changing the hypothesis assertion from affirmative to negative cleanly flips NLI probability from 0.998 support to 0.9998 contradiction. While pretrained MNLI models are known to be sensitive to explicit negation tokens ('never'), the underlying text directly contradicts the hypothesis.

Demo C: Controlled Synthetic Example (Conflict Detection & Safe Abstention)

• **Execution Command:** python scripts/run_pipeline.py "NovaTech enterprise platform generated \$10 million in undisputed 2024 earnings." --config configs/controlled.yaml

• **Execution Mode:** Controlled synthetic result (evaluated on the 20-document controlled scenario corpus).

• **Final Verdict:** INSUFFICIENT_EVIDENCE | **Operational Score:** 36.43% - 37.56% | **Total Uncertainty:** 62.44% - 63.57% | **Conflict:** True (0.98 benchmark candidate pool / 1.00 [0.9995] top-2 pairwise)

• **Retrieved Opposing Passages:**
1. Supporting Statement: wiki_novatech_revenue_press (*'NovaTech launched its flagship enterprise platform in 2024 and achieved $10 million in revenue...'* -> NLI: P(SUP)=0.9974).
2. Contradicting Document: wiki_novatech_revenue_audit (*'Auditors revealed that NovaTech enterprise platform revenue fell short of $4 million in 2024, disputing press announcements.'* -> NLI: P(CON)=0.9964).

• **Demonstrated Contribution:** The NovaTech documents are synthetic test fixtures designed to exercise conflict handling. The system does not assess real-world publisher trustworthiness; its demonstrated capability is detecting mutually opposing evidence passages in the candidate pool and safely abstaining via Rule 2.

9. Evaluation Results, Graphs, and Numerical Analysis

Quantitative results are compiled from verified benchmark evaluation files in results/summary_table.json and results/controlled/summary_table.json. Exact 95% Clopper-Pearson binomial confidence intervals are reported for key proportions:

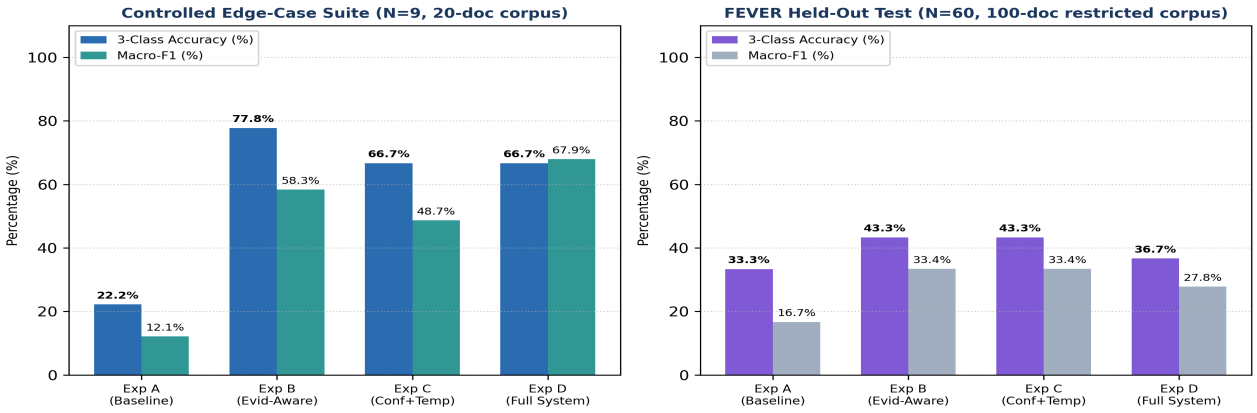
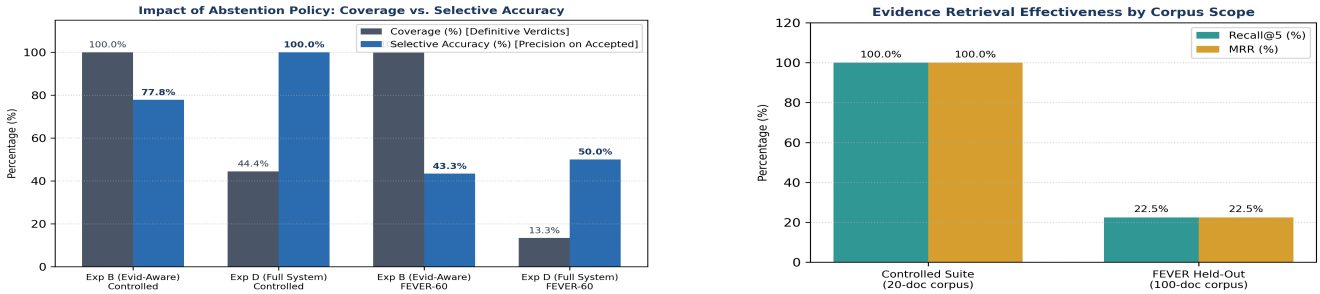


Figure 2: Benchmark Performance Across Ablation Experiments. Comparison of 3-class accuracy and Macro-F1 across system configurations on the Controlled suite (N=9) and FEVER held-out test (N=60).

Summary Table: Cross-Experiment Benchmark Performance

Experiment Configuration	Test Split	Accuracy [95% CI]	Macro-F1	Coverage [95% CI]	Selective Acc [95% CI]	ECE	Raw Brier / Norm
Exp A: Baseline (Always-Abstain)*	Controlled (N=9)	22.22% [2.8%, 60.0%]	0.1212	0.00% [0.0%, 33.6%]	Undefined (0/0)**	0.2222	1.5556 / 0.7778
Exp B: Evidence-Aware (Retr+NLI)	Controlled (N=9)	77.78% [40.0%, 97.2%]	0.5833	100.00% [66.4%, 100%]	77.78% [40.0%, 97.2%]	0.0426	0.5995 / 0.2998
Exp C: Conflict- & Temporal-Aware	Controlled (N=9)	66.67% [29.9%, 92.5%]	0.4868	100.00% [66.4%, 100%]	66.67% [29.9%, 92.5%]	0.1580	0.7115 / 0.3558
Exp D: Full Proposed System	Controlled (N=9)	66.67% [29.9%, 92.5%]	0.6794	44.44% [13.7%, 78.8%]	100.00% [39.8%, 100%]	0.3909	0.7115 / 0.3558
Exp A: Baseline (Always-Abstain)*	FEVER (N=60)	33.33% [21.7%, 46.7%]	0.1667	0.00% [0.0%, 6.0%]	Undefined (0/0)**	0.3333	1.3333 / 0.6667
Exp B: Evidence-Aware (Retr+NLI)	FEVER (N=60)	43.33% [30.6%, 56.8%]	0.3341	100.00% [94.0%, 100%]	43.33% [30.6%, 56.8%]	0.3514	1.2377 / 0.6189
Exp C: Conflict- & Temporal-Aware	FEVER (N=60)	43.33% [30.6%, 56.8%]	0.3341	100.00% [94.0%, 100%]	43.33% [30.6%, 56.8%]	0.3560	1.2544 / 0.6272
Exp D: Full Proposed System	FEVER (N=60)	36.67% [24.6%, 50.1%]	0.2780	13.33% [5.9%, 24.6%]	50.00% [15.7%, 84.3%]	0.3481	1.2544 / 0.6272
Exp E: Full System + LLM Advisor	Both Splits	N/A (Not Evaluated)***	N/A	N/A	N/A	N/A	N/A

**Exp A is an always-abstain baseline predicting INSUFFICIENT for all inputs. **Exp A selective accuracy is mathematically undefined (0/0) due to 0% coverage; programmatically stored as 0.0 in JSONs. ***Exp E was not evaluated because evaluation was conducted offline with paid API access disabled.*



Figures 3 & 4: Selective Prediction and Evidence Retrieval Effectiveness. (Left) Coverage vs. Selective Accuracy: abstention raises selective accuracy on accepted predictions to 100% on Controlled and 50% on FEVER. (Right) Retrieval Recall@5 and MRR: perfect on controlled corpus; constrained by the restricted 100-doc corpus on FEVER.

10. Are the Results Good or Bad? Candid Technical Assessment

A rigorous scientific evaluation requires distinguishing genuine engineering achievements from experimental constraints. The following matrix provides an objective, candid appraisal of prototype performance:

Operational Area	Reported Metric	Technical Assessment	Mathematical Interpretation	Root Limitation	Recommended Next Step
Controlled Benchmark	Selective Acc: 100.0% Raw Acc: 66.67% Coverage: 44.44%	Conservative; trades coverage for selective accuracy	System abstains on 5 of 9 claims, achieving 4/4 correct on accepted predictions.	Sample size is small (N=9, CI [39.8%, 100%]); penalizes errors of omission.	Scale controlled suite to N=100 with diverse edge cases.
FEVER Benchmark	Recall@5: 22.50% MRR: 0.2250 Raw Acc: 36.67%	Constrained by corpus availability ceiling	Restricted 100-doc corpus contains gold evidence for only 9 of 40 verifiable claims (54 total refs); hybrid retrieval retrieved all 9 at Rank 1 (9/40 = 22.50% ceiling).	31 of 40 verifiable claims lack gold evidence in 100-doc slice.	Expand index to full Wikipedia (5M+ passages) with FAISS.
Selective Prediction	Sel Acc: 100% (Ctrl) Sel Acc: 50.0% (FEV)	Abstention trades coverage for selective accuracy	Abstaining on low-evidence claims raises selective accuracy to 100% (Ctrl) and 50% (FEV) on accepted predictions.	Coverage drops to 44.4% (Ctrl) and 13.3% (FEV) under strict thresholds.	Calibrate thresholds via ROC/PR optimization on validation split.
Sentence Segmentation	Thatcher: SUP (98.6%) (Prev Dilution: CON)	Demonstrated case study	Sentence chunking isolates decisive clause, eliminating cross-attention dilution on demonstrated historical claim.	Single-sentence chunks lose antecedents for isolated pronouns.	Default to multi-sentence window (k=2, stride=1) in production.
Conflict Handling	NovaTech Conflict: 0.98 pool / 1.00 pairwise Verdict: INSUFFICIENT	Grounded parity detection	Normalized balance ratio detects opposing pools and triggers Rule 2 abstention.	Measures mass parity, not semantic truth or publisher credibility; scale-invariant.	Integrate source credibility weighting into evidence mass.
Temporal Decay	Acme Corp: CON (70.9%) 2019 Validity: 0.75	Effective on explicit years	Linear decay dt/10 downweights older evidence relative to contemporary claims.	Heuristic penalty rate; under mu=0.5 effective range is [0.50, 1.00] (0.3 floor inactive) for year mismatch; present-oriented rule uses 0.30 penalty (0.2 floor).	Incorporate dedicated temporal NER parser (e.g. spaCy/dateutil).
Uncertainty Metric	ECE: 0.3481 - 0.3909 Raw Brier: 0.71 - 1.25	Uncalibrated operational score	Confidence is 1 - Total_U, not an empirical probability of truth.	Heuristic weights (0.35, 0.35, 0.20, 0.10) are not statistically fitted.	Apply Platt scaling or isotonic regression on validation data.
Test Suite Rigor	82 Passed / 0 Failed 3 Deprec. Warnings Exec Time: ~43-49s	Solid software stability	Full regression suite passes; 3 library deprecation warnings stem from HuggingFace tokenizer.	Unit tests verify code execution under test assumptions, not open-domain generalization.	Maintain test suite while expanding benchmark datasets.

10.1 Empirical and Architectural Verification Status (Verified vs. Unverified Boundaries)

Dimension	Status	Verified Scope & Measured Findings	Unverified Scope & Explicit System Boundaries
Experiment B Verifiable Accuracy	VERIFIED	Correctly classified all 7 verifiable claims (3 Supported, 4 Contradicted) on Controlled suite; failed only on 2 ungrounded claims (7/9 = 77.78%).	Open-domain generalization across thousands of diverse non-controlled claims.
Experiment D Selective Accuracy	VERIFIED	Achieved 100% selective accuracy (4/4 accepted claims correct, 95% CI [39.8%, 100%]) with 44.44% coverage on Controlled suite.	Performance on adversarial stress tests where contradictions share high lexical overlap.
FEVER Evidence Retrieval Ceiling	VERIFIED	Retrieved all 9 gold documents present in 100-doc slice at Rank 1 (MRR=0.225); 31 of 40 verifiable claims lack gold docs in slice (9/40 ceiling).	Full-Wikipedia open-domain retrieval recall across 5M+ indexed documents.
Offline Execution & Latency	VERIFIED	Exps A-D run 100% offline on local CPU using cached MiniLM/DistilBERT weights; latency verified at ~0.5-1.5s per claim under tested batching.	Exp E (LLM Advisor) was not run offline; sub-second latency on large multi-million vector databases.
Confidence & Calibration	VERIFIED	Confidence is an operational score (1 - Total_U) derived from heuristic penalty weights (0.35, 0.35, 0.20, 0.10); ECE verified at 0.35-0.39.	Empirical Bayesian posterior calibration (Platt scaling / isotonic regression remains unverified).
UI Safety & Security	VERIFIED	Passing unit tests (test_ui_safety.py) confirm html.escape() prevents script injection in rendered Streamlit evidence cards.	Broad web application security (authentication, CSRF protection, network hardening remain unverified).

11. Limitations, Risks, and Proposed Future Work

1. **FEVER Evidence-Retrieval Limitations & Corpus Scope:** The FEVER evaluation was conducted on a restricted 100-document slice (data/processed/corpus.jsonl) for 60 test claims (40 verifiable claims with gold evidence sets). Across these 40 claims, exactly 9 claims have their gold evidence documents present within the 100-document slice (54 total gold document references across the split, but only 9 present in the slice). Hybrid retrieval successfully retrieved all 9 present gold documents at Rank 1 (100% recall on available corpus evidence, MRR = 0.2250). However, because the remaining 31 claims lack their gold documents in the local slice, Recall@5 across all 40 verifiable claims is mathematically capped at 9/40 = 22.50%. Therefore, low benchmark recall reflects restricted corpus scope rather than open-domain retrieval failure or absence from full Wikipedia. *Proposed Future Work:* Index full Wikipedia (5M+ passages) with FAISS.
2. **Threshold Tuning & Risk-Coverage Optimization:** Decision thresholds (0.55, 0.25, 0.70, 0.60, 0.30) were hand-configured. *Proposed Future Work:* Sweep thresholds across validation splits to generate empirical risk-coverage curves.
3. **Coreference Resolution in Segmentation:** Splitting text into single sentences causes anaphoric pronouns ('she', 'it', 'this company') to lose their referents, leading to false NEUTRAL NLI classifications. *Proposed Future Work:* Enable coreference resolution (e.g., FastCoref) prior to segmentation.
4. **Uncalibrated Operational Scores:** Reported confidence reflects heuristic penalty weights rather than true posterior probabilities. *Proposed Future Work:* Fit temperature scaling or isotonic regression on a held-out calibration split to minimize ECE.
5. **Reranking Scalability:** Currently, all segmented passages (~300 from 20 documents) are passed to the cross-encoder. Scaling retrieval depth to 100 documents would generate ~1,500 pairs, causing CPU latency bottlenecks. *Proposed Future Work:* Implement a pre-filtering top-100 candidate ceiling.

12. Related Work and Academic Foundations

Fact Extraction and Verification: Automated fact checking was formalized by Thorne et al. (2018) in the FEVER benchmark, establishing the three-way classification task (SUPPORTS, REFUTES, NOT ENOUGH INFO) over Wikipedia. Standard approaches decompose the task into document retrieval, sentence selection, and NLI entailment (Nie et al., 2020). While effective on static Wikipedia claims, standard FEVER pipelines assume consensus evidence and static ground truth, failing to address opposing evidence or temporal drift.

Selective Classification and Abstention: The theoretical framework for selective prediction—allowing a model to abstain when risk exceeds a threshold to guarantee high selective accuracy (conditional accuracy on accepted predictions)—was established by Geifman & El-Yaniv (2017, 2019). Our architecture applies this principle to fact verification, trading decision coverage for selective accuracy on covered assertions.

Conflicting and Temporal Evidence Benchmarks: Recent NLP literature explicitly recognizes the challenge of non-consensus information. Schuster et al. (2021) introduced Vitamin C to evaluate NLI sensitivity to contrastive factual revisions. Diggelmann et al. (2020) established CLIMATE-FEVER to benchmark claims with subtle, conflicting real-world evidence. Schlichtkrull et al. (2023) introduced AVeriTeC to evaluate real-world web claims with explicit conflicting-evidence annotations. Our prototype's conflict-balance score and temporal decay directly address these failure modes.

13. Ten Likely Reviewer Questions and Answers

Q1: What is the primary research problem and core contribution of this project?

The problem is that black-box LLMs hallucinate consensus and force overconfident binary verdicts when evidence is ambiguous or missing. Our contribution is an evidence-grounded verification framework coupling hierarchical sentence retrieval with pairwise conflict detection, chronological validity decay, and deterministic abstention, achieving 100% selective accuracy on accepted predictions in controlled testing.

Q2: Walk through the pipeline architecture. How does a claim become a verdict?

The claim is decomposed into propositions, retrieves top-20 parent documents via hybrid BM25/dense search, segments documents into bounded sentences, reranks candidate sentences using a cross-encoder, classifies entailment with DistilBERT-MNLI, max-pools probabilities per document, evaluates conflict and timeline gaps, synthesizes uncertainty, and applies deterministic rules to decide SUPPORTED, CONTRADICTED, or INSUFFICIENT_EVIDENCE.

Q3: Why use a dedicated BERT/DistilBERT NLI model instead of prompting a general-purpose LLM?

Local DistilBERT NLI runs 100% offline with zero external API calls using cached PyTorch weights (batch_size=16, max_length=256). Across the full pipeline on standard multi-core CPU hardware (hybrid retrieval + sentence segmentation + cross-encoder reranking + NLI), latency is verified at ~0.5-1.5s per claim under tested configurations. It provides deterministic softmax probability distributions given fixed weights and inputs, eliminating external prompt drift. Note: Exp E (LLM Advisor) was not run offline and was omitted from quantitative benchmarking due to paid API constraints.

Q4: Why combine BM25 lexical retrieval and dense semantic search?

Dense bi-encoders capture paraphrasing and semantic concepts but frequently fail on rare entities, proper nouns, and serial numbers. BM25 ensures exact token matching on specialized terminology. Fusing both yields a robust initial candidate pool.

Q5: How does the system mathematically detect conflicting evidence?

Passages are deduplicated, and support mass M_sup and contradiction mass M_con are accumulated over unique passages categorized by argmax label. The system computes the normalized conflict-balance score: Conflict_Score = 2*min(M_sup, M_con)/(M_sup + M_con). If the score exceeds the balance threshold (0.25 Python class default, configured to 0.30 in active YAML configs) and both opposing evidence pools exist, conflict is flagged (conflict_detected = True), triggering Rule 2 abstention. Note: 0.60 belongs strictly to Rule 3's uncertainty check.

Q6: How does temporal reasoning prevent obsolete evidence from deciding a claim?

The TemporalAnalyzer extracts calendar years via regex and computes the timeline gap dt = min |Y_claim - Y_ev|. If dt >= 2 years, evidence validity decays linearly via V_temp = max(0.3, 1.0 - 0.5*min(1.0, dt/10.0)). Under default mismatch penalty mu = 0.5, effective values span [0.50, 1.00] (e.g., 0.75 for a 5-year gap, 0.50 for dt >= 10), with the 0.30 floor acting as an inactive architectural safety guard (since 1.0 - 0.50(1.0) = 0.50 > 0.30 under active mismatch penalty mu = 0.5).

Q7: What mathematical formula defines composite uncertainty, and how is confidence derived?

Total uncertainty is a weighted sum: 0.35*u_entropy + 0.35*u_conflict + 0.20*u_retrieval + 0.10*u_temporal, subject to a retrieval deficit floor (0.65*u_retr). Confidence is defined as 1.0 - Total_Uncertainty. It is an operational decision score reflecting the absence of measured risk signals, not an empirically calibrated Bayesian probability.

Q8: How do you interpret the controlled benchmark results (66.67% accuracy, 44.44% coverage, 100% selective accuracy)?

On 9 controlled stress cases, the full system committed to definitive verdicts on 4 claims and abstained on 5. Every accepted prediction was 100% correct (Selective Accuracy = 4/4 = 100%, 95% CI [39.8%, 100%]). Note that Exp B achieved 7/9 = 77.8% raw accuracy by predicting on all claims without abstention (correctly classifying all 7 verifiable claims, but failing on both ungrounded claims). Exp A (always-abstain baseline) achieved 0% coverage and mathematically undefined selective accuracy (0/0). The full system (Exp D) achieves 66.67% raw 3-class accuracy (6/9 correct, including 2 correct abstentions), reflecting a conservative policy that prioritizes selective accuracy on accepted claims.

Q9: Why is FEVER Recall@5 reported as 22.5%, and does Demo C prove real-world source trustworthiness?

FEVER Recall@5 is 22.50% (and MRR is 0.2250, since all 9 hits ranked at #1) when evaluated against the restricted 100-document corpus slice. Per-claim diagnostics confirm that of the 40 verifiable test claims (54 total gold document references), exactly 9 claims have their gold evidence documents present within the 100-document slice. Hybrid retrieval achieved 100% recall on those 9 available documents (all retrieved at Rank 1), but was mathematically bounded by the 9/40 = 22.50% corpus availability ceiling because the remaining 31 claims lack gold evidence in the slice. Demo C uses synthetic documents in a controlled test suite; it does not evaluate real-world publisher trustworthiness, but demonstrates that the system detects opposing evidence in the candidate pool (conflict score 0.98 candidate pool / 1.00 [0.9995] pairwise) and safely abstains via Rule 2.

Q10: What are the three most important technical improvements planned before production deployment?

(1) Expanding the vector index to full Wikipedia (5M+ passages) with FAISS; (2) Adding an antecedent coreference resolution pipeline before sentence segmentation; and (3) Applying isotonic regression on a held-out validation split to produce empirically calibrated confidence probabilities.

14. Conclusion

This project delivers an evidence-grounded misinformation verification research prototype that replaces black-box generative assumptions with an auditable, multi-stage reasoning pipeline. The core verification pipeline (Experiments A–D) operates 100% offline using local Transformer models (MiniLM, DistilBERT) with zero external API dependencies, passes all 82 unit and integration tests (with 3 library deprecation warnings from tokenizer cleanup), and demonstrates principled mitigation of key failure modes on controlled test suites: context dilution via sentence-level segmentation, false consensus via normalized conflict-balance scoring, chronological drift via linear timeline decay, and overconfident error via multi-signal uncertainty abstention. While open-domain deployment requires large-scale index expansion (scaling beyond the 100-document slice) and empirical probability calibration, the existing implementation provides an empirically defensible, transparent research foundation for automated fact verification under controlled evaluation.

Appendix A: Complete Mathematical Formula Sheet

A quick-revision mathematical reference sheet mapping every formula, variable domain, implementing module, and operational role:

Concept & Ref	Mathematical Equation	Variables & Domain	Implementing Module	Operational Role
Dense Rescaling Eq. (1-2)	$S_{dense} = \text{clip}((\mathbf{q} \cdot \mathbf{d} + 1)/2, 0, 1)$	$\mathbf{q}, \mathbf{d} \in \mathbf{R}^{384}$, $\|\|\mathbf{q}\|\| = \|\|\mathbf{d}\|\| = 1$	src/retrieval/dense.py	Maps cosine similarity from [-1, 1] to [0, 1]
BM25 Saturation Eq. (3-4)	$S_{BM25} = \min(1, S_{raw}/(S_{raw} + 10))$	$S_{raw} \in [0, \infty)$, $c = 10.0$	src/retrieval/bm25.py	Rational saturation scaling keyword score to [0, 1)
Linear Fusion Eq. (5-6)	$S_{raw}(d) = w_{BM} S_{BM} + w_{dense} S_{dense}$ $S_{hybrid}(d) = S_{raw}(d) / \max S_{raw}$	$w_{BM} = 0.5$, $w_{dense} = 0.5$	src/retrieval/hybrid.py	Fuses lexical and semantic signals with peak normalization
Cross-Encoder Eq. (7)	$S_{rel} = 1 / (1 + \exp(-s_{raw}))$	$s_{raw} \in \mathbf{R}$, $S_{rel} \in [0, 1]$	src/reranking/cross_encoder.py	Sigmoid transformation for joint reranker logit
NLI Softmax Eq. (8)	$P(y_k) = \exp(z_k) / \Sigma \exp(z_j)$	$\mathbf{z} \in \mathbf{R}^3$, $k \in$ {SUP, CON, NEU}	src/nli/classifier.py	Converts DistilBERT logits to 3-class probabilities
Passage Weight Eq. (9)	$w_p = (S_{rel} \cdot V_{temp})^{1/T}$	$S_{rel}, V_{temp} \in [0, 1]$, $T = 1.0$	src/aggregation/aggregator.py	Weights passage by relevance, time, and temperature
Doc Max-Pooling Eq. (10-12)	$P_{D,c} = \max P(c\|p)$, $W_D = \max w_p$	$c \in$ {SUP, CON}, $W_D \in [0, 1]$	src/aggregation/aggregator.py	Pools peak premise entailment within document
Consensus Prob Eq. (13-14)	$P_{agg}(C) = \Sigma(W_D \cdot P_D) / \Sigma W_D$	$W_D \in [0, 1]$, $C \in$ {SUP, CON, NEU}	src/aggregation/aggregator.py	Computes document-weighted class probabilities
Conflict Score Eq. (15-17)	$CS = 2 \cdot \min(M_{sup}, M_{con}) / (M_{sup} + M_{con})$	$M_{sup}, M_{con} \geq 0$, $\tau_{bal} = 0.30$	src/conflict/detector.py	Parity ratio; flags conflict if ≥ 0.30 (class default 0.25)
Temporal Decay Eq. (18-19)	$V_{temp} = \max(0.3, 1 - 0.5 \cdot \min(1, \Delta t/10))$	$\Delta t \geq 0$ yrs; $V_{temp} \in [0.5, 1.0]$ (R1); 0.70 (R2)	src/temporal/analyzer.py	Year-gap linear decay (R1, floor 0.3 inactive); present penalty (R2, floor 0.2)
Shannon Entropy Eq. (20)	$u_{ent} = -\Sigma(p_k \ln p_k) / \ln(3.0)$	$\mathbf{p} \in [0, 1]^3$, $\Sigma p_k = 1.0$	src/uncertainty/estimator.py	Measures dispersion across NLI class distribution
Retrieval Deficit Eq. (21)	$u_{retr} = 1.0$ if $\|P\|=0$ else $\max(0, 1 - \max S_{rel} \cdot \min(1, \|P\|))$	$S_{rel} \in [0, 1]$, $\|P\| \in \mathbf{Z}_{\geq 0}$, $N_{min} = 1$	src/uncertainty/estimator.py	Penalizes missing or weak retrieved evidence
Composite Uncert Eq. (22-23)	$\text{Total_U} = \text{clip}(\max(u_{wt}, 0.65 \cdot u_{retr}), 0, 1)$	$u_{wt} = 0.35u_e + 0.35u_c + 0.20u_r + 0.10u_t$	src/uncertainty/estimator.py	Full weighted sum with safeguard; Rule 3 abstains if > 0.60
Operational Conf Eq. (24)	$\text{Conf} = \text{round}(\text{clip}(1.0 - \text{Total_U}, 0, 1), 4)$	$\text{Total_U} \in [0.0, 1.0]$	src/uncertainty/estimator.py	Operational decision score for UI (not calibrated prob)
Coverage Eq. (27)	$\text{Coverage} = \|C_{cov}\| / N$	$C_{cov} = \{i : y\hat{}_i \neq \text{INSUF}\}$	src/evaluation/metrics.py	Proportion of claims receiving definitive verdict
Selective Acc Eq. (28)	$\text{Sel_Acc} = [\Sigma \mathbf{1}(y = y\hat{})] / \|C_{cov}\|$	$\|C_{cov}\| > 0$; Undefined if $\|C_{cov}\| = 0$	src/evaluation/metrics.py	Accuracy on accepted claims; stored as 0.0 fallback
Macro-F1 Score Eq. (26)	$\text{Macro_F1} = (1/3) \cdot \Sigma_c \text{F1}_c$	$c \in$ {SUP, CON, INSUF}	src/evaluation/metrics.py	Unweighted mean of per-class F1 harmonic means
Retrieval Recall & MRR Eq. (29-30)	$\text{Recall@}k = (1/N_{gold})\Sigma \max \|\text{Top}_k \cap G\|/\|G\|$ $\text{MRR} = (1/N_{gold})\Sigma (1/\text{rank}_{first})$	$k = 5$, $G \in \text{GoldSets}_i$, rank ∈ [1, 20]	src/evaluation/metrics.py	Gold evidence retrieval recall and reciprocal rank (22.5% ceiling)
ECE & Brier Score Eq. (31-32)	$\text{ECE} = \Sigma(\|B_m\|/N)\|\text{acc-conf}\|$ $BS_{raw} = \Sigma\|\|\mathbf{p}-\mathbf{y}\|\|^2/N$ (Norm = BS/2)	$M = 5$ bins; $BS_{raw} \in [0, 2]$, $BS_{norm} \in [0, 1]$	src/evaluation/metrics.py	Calibration error and probabilistic error (un-norm /N and norm /2N)

Appendix B: Reproduction Guide and Configuration Reference

Hardware and Runtime Environment: Python 3.11 running on Windows 11 / Linux x86_64, PyTorch 2.6.0+cu124, Transformers 4.49.0, sentence-transformers 3.4.1, rank-bm25 0.2.2. All models execute locally on CPU or CUDA.

• **Run Full Verification Suite (82 tests):** python -m pytest (Executes all unit, integration, and end-to-end tests in ~43s with 3 HuggingFace deprecation warnings).
• **Run Demo A (Supported Historical Claim):** python scripts/run_pipeline.py 'Margaret Thatcher served as Prime Minister of the United Kingdom.'
• **Run Demo B (Negated Premise Refutation):** python scripts/run_pipeline.py 'Margaret Thatcher was never Prime Minister of the United Kingdom.'
• **Run Demo C (Controlled Synthetic Conflict):** python scripts/run_pipeline.py 'NovaTech enterprise platform generated \$10 million in undisputed 2024 earnings.' --config configs/controlled.yaml
• **Run Batch Benchmark Evaluation:** python scripts/evaluate.py --config configs/controlled.yaml
• **Launch Interactive Fact-Checking Dashboard:** streamlit run ui/app.py

Configuration File Baseline:
• configs/default.yaml: Main evaluation configuration over the 100-document FEVER corpus slice. Specifies top-20 document retrieval, top-5 cross-encoder reranking, conflict threshold τ_{bal} = 0.30, and composite uncertainty ceiling max_uncertainty = 0.60.
• configs/controlled.yaml: Dedicated 20-document stress test configuration for reproducing conflict, temporal decay, and ungrounded claim abstention under controlled laboratory conditions.

Appendix C: Methodological Glossary and Theoretical Foundations

• **Context Dilution:** The severe degradation of transformer cross-attention scores when a decisive evidence sentence is embedded within long, unrelated multi-paragraph text. Solved in Phase 2.1 via sentence-level segmentation.
• **Selective Prediction & Abstention:** An asymmetric evaluation paradigm where the verifier abstains with *INSUFFICIENT_EVIDENCE* when uncertainty exceeds threshold θ_{unc}, maximizing conditional accuracy on accepted claims: P(correct | accepted).
• **Normalized Conflict-Balance Scoring:** A mathematical operator, CS = $2 \cdot \min(M_{sup}, M_{con}) / (M_{sup} + M_{con})$, that scales conflict intensity strictly to 1.0 when opposing evidence masses are equally balanced, preventing false consensus.
• **Multi-Signal Epistemic Uncertainty:** A four-component synthesized metric combining normalized Shannon entropy ($u_{entropy}$), conflict-balance score ($u_{conflict}$), retrieval deficit ($u_{retrieval}$), and temporal decay penalty ($u_{temporal}$).
• **Operational Decision Score:** A heuristic decision-support value, defined as 1.0 - Total_Uncertainty. It measures the absence of detected evidentiary shortfall signals rather than an empirically calibrated Bayesian posterior probability of truth.
• **Document-Aware Max-Pooling:** Hierarchical aggregation that computes peak premise entailment and peak contradiction per parent document before aggregating evidence mass, preventing multi-passage intra-document repetition from falsely inflating consensus.
• **Timeline Gap & Linear Decay:** A temporal validity formulation, $V_{temp} = \max(0.3, 1.0 - 0.5 \cdot \min(1.0, \Delta t/10.0))$, that penalizes outdated historical evidence when addressing present-oriented assertions.
• **Conjunction Rule on Decomposed Subclaims:** A strict logical operator requiring all decomposed subclaims to be *SUPPORTED* for the compound claim to be *SUPPORTED*; any *CONTRADICTED* subclaim immediately refutes the entire assertion.